

Appendix L

Miscellaneous methods

L.1 Higher-order integration schemes

The basic idea behind the predictor-corrector algorithms is to use information about the position and its first n derivatives at time t to arrive at a prediction for the position and its first n derivatives at time $t + \Delta t$. We then compute the forces (and thereby the accelerations) at the predicted positions. And then we find that these accelerations are *not* equal to the values that we had predicted. So we adjust our predictions for the accelerations to match the facts. But we do more than that. On the basis of the observed discrepancy between the predicted and observed accelerations, we also try to improve our estimate of the positions and the remaining $n - 1$ derivatives. This is the “corrector” part of the predictor-corrector algorithm. The precise “recipe” used in applying this correction is a compromise between accuracy and stability. Here, we shall simply show a specific example of a predictor-corrector algorithm, without attempting to justify the form of the corrector part.

Consider the Taylor expansion of the coordinate of a given particle at time $t + \Delta t$:

$$r(t + \Delta t) = r(t) + \Delta t \frac{\partial r}{\partial t} + \frac{\Delta t^2}{2!} \frac{\partial^2 r}{\partial t^2} + \frac{\Delta t^3}{3!} \frac{\partial^3 r}{\partial t^3} + \cdots .$$

Using the notation

$$\begin{aligned} x_0(t) &\equiv r(t) \\ x_1(t) &\equiv \Delta t \frac{\partial r}{\partial t} \\ x_2(t) &\equiv \frac{\Delta t^2}{2!} \frac{\partial^2 r}{\partial t^2} \\ x_3(t) &\equiv \frac{\Delta t^3}{3!} \frac{\partial^3 r}{\partial t^3}, \end{aligned}$$

we can write the following *predictions* for $x_0(t + \Delta t)$ through $x_3(t + \Delta t)$:

$$\begin{aligned} x_0(t + \Delta t) &= x_0(t) + x_1(t) + x_2(t) + x_3(t) \\ x_1(t + \Delta t) &= x_1(t) + 2x_2(t) + 3x_3(t) \\ x_2(t + \Delta t) &= x_2(t) + 3x_3(t) \\ x_3(t + \Delta t) &= x_3(t). \end{aligned}$$

Now that we have $x_0(t + \Delta t)$, we can compute the forces at the predicted position, and thus compute the corrected value for $x_2(t + \Delta t)$. We denote the difference between $x_2^{\text{corrected}}$ and $x_2^{\text{predicted}}$ by Δx_2 :

$$\Delta x_2 \equiv x_2^{\text{corrected}} - x_2^{\text{predicted}}.$$

We now estimate “corrected” values for x_0 through x_3 , as follows:

$$x_n^{\text{corrected}} = x_n^{\text{predicted}} + C_n \Delta x_2, \quad (\text{L.1.1})$$

where the C_n are constants fixed for a given order algorithm. As indicated, the values for C_n are such that they yield an optimal compromise between the accuracy and the stability of the algorithm. For instance, for a fifth-order predictor-corrector algorithm (i.e., one that uses x_0 through x_4), the values for C_n are

$$\begin{aligned} C_0 &= \frac{19}{120} \\ C_1 &= \frac{3}{4} \\ C_2 &= 1 && (\text{of course}) \\ C_3 &= \frac{1}{2} \\ C_4 &= \frac{1}{12}. \end{aligned}$$

One may iterate the predictor and corrector steps to self-consistency. However, there is little point in doing so because (1) every iteration requires a force calculation. One would be better off spending the same computer time to run with a *shorter* time step and only one iteration because (2) even if we iterate the predictor-corrector algorithm to convergence, we still do not get the *exact* trajectory: the error is still of order Δt^n for an n th-order algorithm. This is why we gain more accuracy by going to a shorter time step than by iterating to convergence at a fixed value of Δt .

L.2 Surface tension via the pressure tensor

If we have an interface in our system we can compute the interfacial tension, γ , from the pressure tensor. In a homogeneous system at equilibrium, the thermodynamic pressure is constant and equal in all directions. For an inhomogeneous system, mechanical equilibrium requires that the component of the pressure tensor normal to the interface is constant throughout the system. The components tangential to the interface can vary in the interfacial region, but must be equal to the normal component in the bulk liquids.

For an inhomogeneous fluid there is no unambiguous way to compute the normal p_n and tangential p_t components of the pressure tensor [727–729]. Here, we have used the Kirkwood-Buff convention [730] for expressing the stress tensor in a system of particles with pairwise-additive interactions. The system is divided into N_{sl} equal slabs parallel to the x, y plane. The local normal ($p_n(k)$) and tangential ($p_t(k)$) components of the pressure tensor are given by [143]

$$p_n(k) = k_B T \langle \rho(k) \rangle - \frac{1}{V_{sl}} \left\langle \sum_{(i,j)}^{(k)} \frac{z_{ij}^2}{r_{ij}} \frac{dU(r_{ij})}{dr} \right\rangle, \quad (\text{L.2.1})$$

and

$$p_t(k) = k_B T \langle \rho(k) \rangle - \frac{1}{2V_{sl}} \left\langle \sum_{(i,j)}^{(k)} \frac{x_{ij}^2 + y_{ij}^2}{r_{ij}} \frac{dU(r_{ij})}{dr} \right\rangle, \quad (\text{L.2.2})$$

where $\langle \rho(k) \rangle$ is the average density in slab k , $V_{sl} = L_x L_y L_z / N_{sl}$ is the volume of a slab, $U(r)$ is the intermolecular potential from which the conservative forces can be derived. $\sum_{(i,j)}^{(k)}$ means that the summation runs over all pairs of particles i, j for which the slab k (partially) contains the line that connects the particles i and j . Slab k gets a contribution $1/N_o$ from a given pair (i, j) , where N_o is the total number of slabs which intersect this line.

It can be shown that, even though the stress tensor is not unique, the definition of the interfacial tension γ is free from ambiguities [729]. The interfacial tension can be calculated by integrating the difference between the normal and tangential components of the pressure tensor across the interface. In the case of our system with two interfaces, γ reads

$$\gamma = \frac{1}{2} \int_o^{L_z} dz [p_n(z) - p_t(z)]. \quad (\text{L.2.3})$$

The factor $\frac{1}{2}$ corrects for the fact that, in a system with periodic boundary conditions, interfaces necessarily come in pairs.

L.3 Micro-canonical Monte Carlo

Most experimental observations are performed at constant N, P, T ; sometimes at constant μ, V, T ; and occasionally at constant N, V, T .

Experiments at constant N, V, E exist (e.g., in adiabatic calorimetry) but are rare, and it is fair to say that microcanonical MC simulations of dense liquids or solids are equally rare. In fact, the first microcanonical Monte Carlo method was suggested by Creutz [172], in the context of lattice gauge theory MC simulations. The microcanonical Monte Carlo method does not use random numbers to determine the acceptance of a move, but it does use random numbers to generate trial moves.

The constant NVE MC algorithm uses the following procedure. We start with the system in a configuration $\mathbf{q}^N$. We denote the potential energy for this state by $U(\mathbf{q}^N)$. We now fix the total energy of the system at a value $E > U$. The excess energy E_D (the D stands for “demon”) is equal to $E - U$ and is carried by an additional degree of freedom. E_D must be non-negative. Now we start our Monte Carlo run.

1. After each trial move from an “old” configuration (o) to a trial configuration (n), we compute the change in potential energy of the system,

$$\Delta U = U(\mathbf{q}^N(n)) - U(\mathbf{q}^N(o)).$$

2. If $\Delta U < 0$, we accept the move and increase the energy carried by the demon by $|\Delta U|$. If $\Delta U > 0$, we test if the demon carries enough energy to make up the difference. Otherwise, we reject the trial move.

Note that no random numbers were used in this decision. Once the system has equilibrated, the probability density to find the demon with an energy E_D is given by the Boltzmann distribution:

$$\mathcal{N}(E_D) = (k_B T)^{-1} \exp(-E_D/k_B T),$$

where T is the temperature that the system reaches after equilibration. Hence, the demon acts as a thermometer. In that sense, the demon energy plays the same role as the kinetic energy in a micro-canonical MD simulation.

Micro-canonical Monte Carlo is rarely used to simulate molecular systems but it finds many other applications in cases where canonical MC simulations would fail, for instance in the study of systems interacting through gravity, as their potential energy is not bounded from below.

L.4 Details of the Gibbs “ensemble”

The introduction of a new ensemble brings up the question of whether it is a “proper ensemble”; that is, does it yield the same results as the conventional ensembles? To prove it does, we use the partition function (6.6.1) as derived in section 6.6.2 to define a free energy. This free energy is used to show that, in the thermodynamic limit, the Gibbs ensemble and the canonical ensemble are equivalent. This proof gives considerable insight into why the method works. Before we proceed, we first list a few basic results for the free energy in the canonical ensemble.

L.4.1 Free energy of the Gibbs ensemble

L.4.1.1 Basic definitions and results for the canonical ensemble

Consider a system of N particles in a volume V and temperature T (canonical ensemble). The partition function is defined as (see Ruelle [731])

$$Q(N, V, T) \equiv \frac{1}{\Lambda^{3N} N!} \int_V \mathbf{dr}^N \exp[-\beta U(N)]. \quad (\text{L.4.1})$$

The free energy density is defined in the thermodynamic limit by

$$f(\rho) \equiv \lim_{V \rightarrow \infty} f_V(\rho) \equiv \lim_{\substack{V \rightarrow \infty \\ N/V = \rho}} -\frac{1}{\beta V} \ln Q_{N,V,T},$$

where $\rho = N/V$ is the density of the system. For a finite number of particles we can write

$$Q(N, V, T) = \exp \{-\beta V [f(\rho) + o(V)]\}, \quad (\text{L.4.2})$$

where $g(V) = o(V)$ means $g(V)/V$ approaches 0 as $V \rightarrow \infty$. With this free energy, we can derive some interesting properties of a canonical system in the thermodynamic limit.

For example, it can be shown that this free energy is a convex function of the density ρ [731]:

$$f(x\rho_1 + (1-x)\rho_2) \leq xf(\rho_1) + (1-x)f(\rho_2), \quad (\text{L.4.3})$$

for every ρ_1, ρ_2 , and x where $0 \leq x \leq 1$. The equality holds in the case of a first-order transition, if $\rho_g \leq \rho_1 \leq \rho_2 \leq \rho_l$, where ρ_g, ρ_l denote the density of coexisting gas and liquid phases, respectively.

Another interesting result, which plays a central role on the following pages, is the well-known saddle point theorem [732] (also called the *steepest descent method*). This theorem is based on the observation that, for a macroscopic system (N very large) in equilibrium, the probability that the free energy density deviates from its minimum value is extremely small. Therefore, when we calculate for such a system an ensemble average, we have to take into account only those contributions where the free energy has its minimum value. Assume that $Q(N, V, T)$ can be written as

$$Q(N, V, T) \equiv \int da_1, \dots, da_m \exp[-\beta V (f_m(a_1, \dots, a_m) + o(V))],$$

where $a_1, \dots, a_m$ are variables that characterize the thermodynamic state of the system. Furthermore, define

$$f(\rho) \equiv \min_{a_1, \dots, a_m} f_m(a_1, \dots, a_m)$$

and assume that $f_m(a_1, \dots, a_m)$ and the term $o(V)$ satisfy a few technical conditions [732], which hold for most statistical mechanics systems. The saddle point theorem states that, in the thermodynamic limit, the free energy of the system is equal to this minimum value $f(\rho)$ or

$$\lim_{\substack{V \rightarrow \infty \\ N/V = \rho}} -\frac{1}{\beta V} \ln Q(N, V, T) = f(\rho). \quad (\text{L.4.4})$$

Moreover, this saddle point theorem can also be used to calculate the ensemble average of a quantity A :

$$\langle A(a_1, \dots, a_m) \rangle_V \equiv \frac{1}{Q(N, V, T)} \int da_1, \dots, da_m \times A(a_1, \dots, a_m) \exp \{-\beta V [f_m(a_1, \dots, a_m) + o(V)]\}. \quad (\text{L.4.5})$$

In the thermodynamic limit, this ensemble average again has contributions only from those configurations where $f_m(a_1, \dots, a_m)$ has its minimum value. Let us define S as the collection of these minima:

$$S = \left\{ y_1, \dots, y_m \left| f_m(y_1, \dots, y_m) = \min_{a_1, \dots, a_m} f_m(a_1, \dots, a_m) \right. \right\}.$$

We now can state the saddle point theorem in a convenient form by introducing a function $G(a_1, \dots, a_m) \geq 0$ with support on the surface S and normalization

$$\int_S da_1, \dots, da_m G(a_1, \dots, a_m) = 1,$$

such that, for an arbitrary function A ,

$$\begin{aligned} \langle A(a_1, \dots, a_m) \rangle &\equiv \lim_{V \rightarrow \infty} \langle A(a_1, \dots, a_m) \rangle_V \\ &= \int_S da_1, \dots, da_m G(a_1, \dots, a_m) A(a_1, \dots, a_m). \end{aligned} \quad (\text{L.4.6})$$

L.4.1.2 The free energy density in the Gibbs ensemble

The Gibbs ensemble is introduced in section 6.6.2 as an N, V, T ensemble to which an additional degree of freedom is added: the system is divided into two subsystems that have *no* interaction with each other. We can rewrite the partition function of the canonical ensemble (L.4.1):

$$\begin{aligned} Q(N, V, T) &= \frac{1}{\Lambda^{3N} N!} \sum_{n_1=0}^N \binom{N}{n_1} \int_0^V dV_1 \int d\mathbf{r}_1^{n_1} \int d\mathbf{r}_2^{N-n_1} \exp \{-\beta [\mathcal{U}(n_1) \\ &\quad + \mathcal{U}(N - n_1) + \text{interactions between the two volumes}]\}. \end{aligned} \quad (\text{L.4.7})$$

The difference between this equation and the partition function of the Gibbs ensemble (6.6.1) is that, in Eq. (L.4.7), we have interactions between the subsystems. In the case of short-range interactions, the last term in the exponent of Eq. (L.4.7) is proportional to a surface term. This already suggests that both ensembles should behave similarly in many respects. We work out these ideas more rigorously in the following pages.

In the usual way, we define, as a free energy in the Gibbs ensemble,

$$\bar{f}(\rho) \equiv \lim_{\substack{V \rightarrow \infty \\ N/V = \rho}} -\frac{1}{\beta V} \ln \bar{Q}_{N,V,T}. \quad (\text{L.4.8})$$

In the partition function of the Gibbs ensemble (6.6.1), we can substitute Eq. (L.4.1):

$$\bar{Q}(N, V, T) = \sum_{n_1=0}^N \int_0^V dV_1 \, Q(n_1, V_1, T) Q(N - n_1, V - V_1, T).$$

Introducing $x = N_1/N$ and $y = V_1/V$, and assuming that the number of particles is very large, we can then write

$$\bar{Q}(N, V, T) = NV \int_0^1 dx \int_0^1 dy \, \bar{Q}_N(x, y),$$

where

$$\begin{aligned} \bar{Q}_N(x, y) &= Q_{xN, yV, T} Q_{(1-x)N, (1-y)V, T} \\ &= \exp \left\{ -\beta V \left[yf\left(\frac{x}{y}\rho\right) + (1-y)f\left(\frac{1-x}{1-y}\rho\right) + o(V) \right] \right\}. \end{aligned}$$

Note that, in this equation, $f(\rho)$ is the free energy of a canonical system. So, we can apply the saddle point theorem of the previous section (L.4.4) to calculate the free energy density of the Gibbs ensemble $\bar{f}(\rho)$

$$\bar{f}(\rho) = \min_{\substack{0 \leq x \leq 1 \\ 0 \leq y \leq 1}} \left[yf\left(\frac{x}{y}\rho\right) + (1-y)f\left(\frac{1-x}{1-y}\rho\right) \right] \equiv \min_{\substack{0 \leq x \leq 1 \\ 0 \leq y \leq 1}} \bar{f}(x, y).$$

We now have to find the surface S on which the function $\bar{f}(x, y)$ reaches its minimum. For this, we can use that $f(\rho)$ is a convex function of the density (L.4.3). This gives, for $\bar{f}(x, y)$,

$$\bar{f}(x, y) \geq f\left(y\frac{x}{y}\rho + (1-y)\frac{1-x}{1-y}\rho\right) = f(\rho). \quad (\text{L.4.9})$$

We first consider the case where there is only one phase. For this case any combination of x and y that results in densities ρ_1 and ρ_2 in the subsystems different from ρ will give a higher free energy. So, the equality in Eq. (L.4.9) holds only if

$$\frac{x}{y}\rho = \frac{1-x}{1-y}\rho, \quad \text{or} \quad x = y.$$

Thus, when there is only one phase, the free energy of the Gibbs ensemble has its minimum value (in the thermodynamic limit) when both boxes have a density equal to the equilibrium density of the canonical ensemble. Therefore, the surface S is given by

$$S = \{(x, y) | x = y\}.$$

Second, we consider the case of a first-order phase transition. Let ρ be such that $\rho_l \leq \rho \leq \rho_g$, and let us choose x and y such that

$$\rho_g \leq \frac{x}{y}\rho \equiv \rho_3 \leq \rho_l \quad \text{and} \quad \rho_g \leq \frac{1-x}{1-y}\rho \equiv \rho_4 \leq \rho_l. \quad (\text{L.4.10})$$

For this case the equality in Eq. (L.4.3) holds, and we can write, for $\bar{f}(x, y)$,

$$\begin{aligned} \bar{f}(x, y) &= yf(\rho_3) + (1-y)f(\rho_4) \\ &= f(y\rho_3 + (1-y)\rho_4). \end{aligned} \quad (\text{L.4.11})$$

Note that

$$(y\rho_3 + (1-y)\rho_4) = \rho, \quad (\text{L.4.12})$$

which gives

$$\bar{f}(x, y) = f(\rho). \quad (\text{L.4.13})$$

It can be shown that, if x, y do not satisfy Eq. (L.4.10),

$$\bar{f}(x, y) > f(\rho).$$

Therefore, the surface S in the case of a first-order phase transition is given by

$$S = \left\{ (x, y) \left| \rho_g \leq \frac{x}{y}\rho \leq \rho_l, \quad \rho_g \leq \frac{1-x}{1-y}\rho \leq \rho_l \right. \right\}. \quad (\text{L.4.14})$$

This result shows that, in the case of a first-order transition, the (bulk) free energy of the Gibbs ensemble has its minimum value (in the thermodynamic limit) for all values of x, y where there is vapor-liquid coexistence in *both* boxes.

Eqs. (L.4.9) and (L.4.13) show that, in the thermodynamic limit, the free energy of the Gibbs ensemble is equal to the free energy of the canonical ensemble. To calculate an ensemble average, it remains to determine the function $G(x, y)$ using Eq. (L.4.6).

In the case of a pure phase $G(x, y)$ needs to be of the form

$$G(x, y) = g(x)\delta(x - y). \quad (\text{L.4.15})$$

It is shown in the Appendix of [220] that, for an ideal gas, $g(x) = 1$. We expect that the same holds for an interacting gas. Fig. L.1 shows a probability plot in

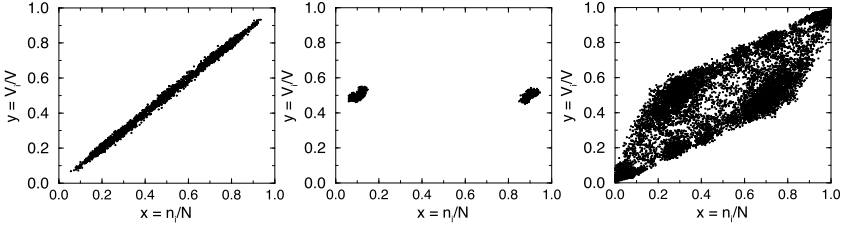


FIGURE L.1 Probability plot in the x, y plane ($x = n_1/N$, $y = V_1/V$ and $x = (N - n_1)/N$, $y = (V - V_1)/V$) for a Lennard-Jones fluid at various temperatures: (left) high temperature ($T = 10$), (middle) well below the critical temperature ($T = 1.15$), and (right) slightly below the critical temperature ($T = 1.30$).

the x, y plane for a simulation of a finite system at high temperature. This figure shows that $x \approx y$.

In the case of two phases, we will show that the system will split up into a liquid phase, with density, ρ_l , in one box, and a vapor phase, with density, ρ_g , in the other box.

Until now we have ignored surface effects, which arise from the presence of a liquid-vapor interface in the boxes. When the density in one box is between the vapor and liquid density the system will form droplets of gas or liquid. The interfacial free energy associated with these droplets has (in the thermodynamic limit) a negligible contribution to the bulk free energy of the Gibbs ensemble. Nevertheless, this surface-free energy is the driving force that causes the system to separate into a homogeneous liquid in one box and a homogeneous vapor phase in the other.

These surface effects are taken into account in the next significant term in the expression for the free energy (L.4.2), which is the term due to the surface tension. This gives, for the partition function,

$$Q(N, V, T) = \exp \{-\beta [Vf(\rho) + \gamma A + o(A)]\}, \quad (\text{L.4.16})$$

where A denotes the area of the interface and γ denotes the interfacial tension. For three-dimensional systems, in general this area will be proportional to $V^{2/3}$. Using this form of the partition function for the Gibbs ensemble, Eq. (L.4.5) can be written as

$$\begin{aligned} \langle A(x, y) \rangle_V & \quad (\text{L.4.17}) \\ &= \frac{\iint dx dy A(x, y) \exp \{-\beta [Vf(x, y) + \gamma V^{2/3}a(x, y) + o(V^{2/3})]\}}{Q(N, V, T)}, \end{aligned}$$

where $a(x, y)$ is a function of the order of unity.

We know from the saddle point theorem that the most important contribution to the integrals comes from the region S , defined by Eq. (L.4.14). Thus,

$$\begin{aligned}
\langle A(x, y) \rangle_V &\approx \frac{\iint_S dx dy A(x, y) \exp \left\{ -\beta \left[V f(x, y) + \gamma V^{2/3} a(x, y) + o(V^{2/3}) \right] \right\}}{\iint_S dx dy \exp \left\{ -\beta \left[V f(x, y) + \gamma V^{2/3} a(x, y) + o(V^{2/3}) \right] \right\}} \\
&= \frac{\iint_S dx dy A(x, y) \exp \left\{ -\beta \left[\gamma V^{2/3} a(x, y) + o(V^{2/3}) \right] \right\}}{\iint_S dx dy \exp \left\{ -\beta \left[\gamma V^{2/3} a(x, y) + o(V^{2/3}) \right] \right\}} \quad (\text{L.4.18})
\end{aligned}$$

and applying the saddle point theorem again

$$\langle A(x, y) \rangle_V \approx \frac{\iint_{S_A} dx dy A(x, y) \exp \left[-\beta \gamma V^{2/3} a(x, y) + o(V^{2/3}) \right]}{\iint_{S_A} dx dy \exp \left[-\beta \gamma V^{2/3} a(x, y) + o(V^{2/3}) \right]} \quad (\text{L.4.19})$$

and

$$\lim_{V \rightarrow \infty} \langle A(x, y) \rangle_V = \iint_{S_A} dx dy G(x, y) A(x, y), \quad (\text{L.4.20})$$

where the surface S_A is now given by

$$S_A = \left\{ (x, y) \left| a(x, y) = \min_{\bar{x}, \bar{y}} a(\bar{x}, \bar{y}) \right. \right\}. \quad (\text{L.4.21})$$

In the infinite system it is easily seen that the area of the interface is 0, if box 1 contains only gas (liquid) and box 2 only liquid (gas). Therefore, the surface S_A contains only two points, which correspond to the vapor and liquid densities:

$$S_A = \left\{ (x, y) \left| \frac{x}{y} = \rho_l \text{ and } \frac{1-x}{1-y} = \rho_g \text{ or } \frac{x}{y} = \rho_g \text{ and } \frac{1-x}{1-y} = \rho_l \right. \right\}. \quad (\text{L.4.22})$$

It is straightforward to show that this surface gives, for $G(x, y)$,

$$\begin{aligned}
G(x, y) &= \frac{1}{2} \delta \left(x - \frac{\rho_g}{\rho} \frac{\rho - \rho_g}{\rho_l - \rho_g} \right) \delta \left(y - \frac{\rho - \rho_g}{\rho_l - \rho_g} \right) \\
&\quad + \frac{1}{2} \delta \left(x - \frac{\rho_l}{\rho} \frac{\rho_l - \rho}{\rho_l - \rho_g} \right) \delta \left(y - \frac{\rho_l - \rho}{\rho_l - \rho_g} \right). \quad (\text{L.4.23})
\end{aligned}$$

We have shown more formally that the free energy density for the Gibbs ensemble, as defined by Eq. (L.4.8), becomes identical to the free energy density of the canonical ensemble. Furthermore, it is shown that, at high temperatures, $x = y$; that is, the densities in the two subsystems of the Gibbs ensemble are equal and equal to the density in the canonical ensemble (see Fig. L.1).

In the case of a first-order phase transition, if surface terms would be unimportant, then x and y are restricted to the area defined by Eq. (L.4.14):

$$\rho_g \leq \frac{x}{y} \rho \equiv \rho_3 \leq \rho_l \quad \text{and} \quad \rho_g \leq \frac{1-x}{1-y} \rho \equiv \rho_4 \leq \rho_l. \quad (\text{L.4.24})$$

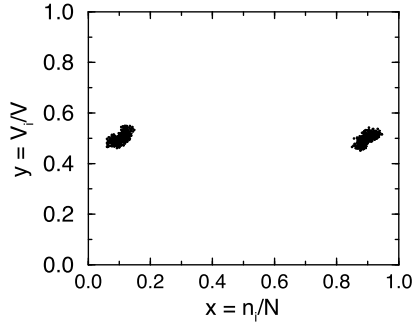


FIGURE L.2 Probability plot in the $x - y$ plane of a successful simulation of a Lennard-Jones fluid well below the critical temperature ($T = 1.15$ and $N = 500$).

If we take surface effects into account, this surface (Eq. (L.4.24)) reduces to two points in the x, y plane. The densities of these points correspond to the density of the gas or liquid phase in the canonical ensemble.

It is interesting to compare this with the results of an actual simulation of a finite system. In Fig. L.1, the results are shown for a simulation at a temperature well below the critical point. Under such conditions, the surface reduces to two points. This should be compared to the results of a simulation close to the critical point (Fig. L.1). Under such conditions the interfacial tension is very small and we see that the simulation samples the entire surface S . Note that due to the finite size of this system, fluctuations are also possible in which the density of a subsystem becomes greater or smaller than the density of the liquid or gas phase.

L.4.2 Graphical analysis of simulation results

In Appendix L.4, we describe a graphical technique for analyzing the results of a Gibbs ensemble simulation. In this scheme, the fraction of all particles (n_i/N) in box i is plotted versus the fraction of the total volume (v_i/V) taken up by this box. In the $x-y$ plane, where $x = n_i/N$ and $y = V_i/V$, every dot represents a point sampled in the simulation.

In the thermodynamic limit, only two points in the $x-y$ plane are sampled; namely, those that correspond to the coexisting liquid and gas density (see Appendix L.4). For a finite system, we expect to observe fluctuations around these points. Fig. L.2 shows an $x-y$ plot for a simulation of two-phase coexistence well below the critical temperature. The fact that the simulation results cluster around the two points that correspond to the coexisting liquid and vapor indicates that the system was well equilibrated. If a simulation in the Gibbs ensemble is performed far below the critical temperature, it is in general no problem to analyze the results. After the equilibration, it becomes clear which of the boxes contains the vapor phase and which the liquid phase. The densities of the coexisting phases can simply be obtained by sampling the densities at regular intervals.

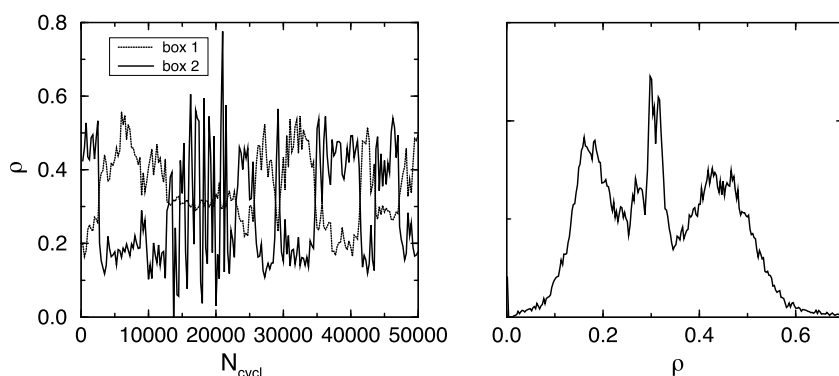


FIGURE L.3 Density in the two boxes in a Gibbs ensemble simulation close to the critical temperature. The left figure shows the evolution of the density of the two boxes during a simulation. The right figure gives the corresponding probability density. The simulations were performed on a Lennard-Jones fluid with $N = 256$ at $T = 1.30$.

When estimating the accuracy of the simulation one should be careful since the “measured” densities are not sampled independently: in estimating the standard deviations of the results one should take this into account (this aspect is discussed in more detail in Appendix A of [733]).

Close to the critical point, however, it is possible that the boxes continuously change “identity” during a simulation. In Fig. L.3 the evolution of the density in such a simulation close to the critical point is shown. In such a system, the average density in any one of the two boxes will tend to the overall density (N/V). In those circumstances, it is more convenient to construct a histogram of the probability density $P(\rho)$ to observe a density ρ in either box. Even when the boxes change identity during the simulation, the maxima of $P(\rho)$ are still well defined. And, as shown in Appendix L.4, in the thermodynamic limit, the two maxima of $P(\rho)$ correspond to coexisting vapor and liquid densities, except precisely at the critical point. (For a discussion of the critical behavior of $P(\rho)$, see the article by Allen and Tildesley [46].) Because $P(\rho)$ is obtained by sampling the density in both boxes, the results are not influenced when the boxes change identity. In Fig. L.3 an example of such a density distribution is shown. In this particular example, the simulation was carried out rather close to the critical point. Under those conditions, the interpretation of the density histogram is complicated because interfaces may form in both boxes. As a consequence, three peaks are observed; the two outside peaks correspond to the coexisting liquid and gas phase. A simple model that accounts for the existence of the middle peak is discussed in [220].

Example 30 (Finite-size effects in the Gibbs ensemble). Most Gibbs-ensemble simulations are performed on relatively small systems ($64 \leq N \leq 500$). One

therefore would expect to see significant finite-size effects, in particular, close to the critical point. Indeed, in simulations of a system of 100 Ising spins on a lattice,^a phase coexistence is observed at temperatures as much as 25% above the critical temperature of the infinite system. In contrast to what is found in lattice gases, the first Gibbs-ensemble studies of the phase diagram of the Lennard-Jones fluid (in two and three dimensions) [77,211,213,220] did not show significant finite-size effects. This striking difference with the lattice models motivated Mon and Binder [734] to investigate the finite-size effects in the Gibbs ensemble for the two-dimensional Ising model in detail. For the two-dimensional Ising model the critical exponents and critical temperature are known exactly. Mon and Binder determined for various system sizes L the order parameter $M_L(T)$ (see Eq. (6.6.13)):

$$M_L(T) = \frac{\rho_l(T) - \rho_c}{\rho_c} = (1 - T/T_c)^\beta,$$

where $\rho_l(T)$ is the density of the liquid phase, ρ_c and T_c are the critical density and temperature, respectively, and β is the critical exponent.

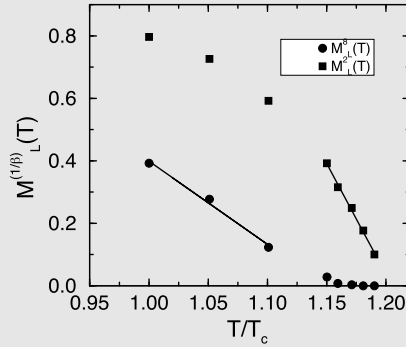


FIGURE L.4 Finite-size effects in a Gibbs-ensemble simulation of the two-dimensional Ising model. Order parameter $M_L^{1/\beta}(T)$ for $L = 10$ (i.e., $L \times L = 100$ spins) versus T/T_c , where T_c is the exact critical temperature for the infinite system. The lines are fitted through the points. The simulation data are taken from [734].

The results of the simulations of Mon and Binder are shown in Fig. L.4, in which the order parameter $M_L(T)$ is plotted as $M_L^{1/\beta}(T)$ versus T/T_c . Such a plot of the order parameter allows us to determine the effective critical exponent of the system. If the system behaves classically, the critical exponent has the mean field value $\beta = 1/2$ and we would expect a linear behavior of $M_L^2(T)$. On the other hand, if the system shows nonclassical behavior, with exponent $\beta = 1/8$, we would expect a straight line for $M_L^8(T)$. Fig. L.4 shows that, away from the critical point, the temperature dependence of the order parameter is best described with an exponent $\beta = 1/8$. Closer to the critical point, the mean field exponent $\beta = 1/2$ fits the data better. This behavior is as expected. Away from the critical point the system can accommodate all rele-

vant fluctuations and exhibits nonclassical behavior. But close to the critical point the system is too small to accommodate all fluctuations and, as a consequence, mean field behavior is observed. In addition, Fig. L.4 shows that we still can observe vapor-liquid coexistence at temperatures 20% above the critical temperature of the infinite system, which implies significant finite-size effects. The study of Mon and Binder shows that, in a lattice model of a fluid, finite-size effects on the liquid-vapor coexistence curve are very pronounced. It is important to note that, in this lattice version of the Gibbs ensemble, we do not change the volume and therefore fewer fluctuations are possible than in the off-lattice version.

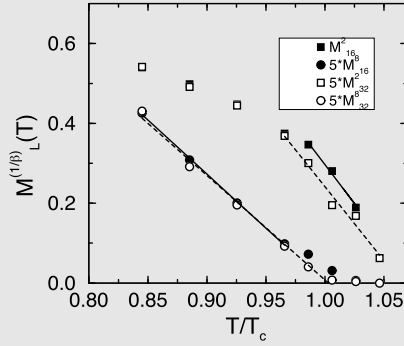


FIGURE L.5 Finite-size effects in the liquid-vapor coexistence curve of the two-dimensional Lennard-Jones fluid (truncated potential $r_c = 5.0\sigma$) studied by Gibbs-ensemble simulation. The order parameter M corresponds to the density difference between the coexisting liquid and vapor phases. The figure shows $M_L^{1/\beta}(T)$ versus T/T_c for various system sizes L . T_c is the estimated critical temperature for the infinite system ($T_c = 0.497 \pm 0.003$). The simulation data are taken from [222].

The striking differences between the findings of Mon and Binder and the results of the early simulations of the Lennard-Jones fluid motivated Panagiotopoulos to reinvestigate in some detail the finite-size effects of Gibbs-ensemble simulations of the two- and three-dimensional Lennard-Jones fluid [222]. The results of the simulations of Panagiotopoulos are shown in Fig. L.5. For the Lennard-Jones fluid, the order parameter is defined as

$$M_L(T) = \rho_l - \rho_g.$$

The results for the two-dimensional Lennard-Jones fluid are qualitatively similar to the results of Mon and Binder. At low temperatures, Ising-like behavior is observed and close to the critical point mean-field-like behavior. An important difference is the magnitude of the finite-size effects. Fig. L.5 shows that, for the two system sizes, the results are very similar; the finite size effects are at most 5%. In addition, Fig. L.5 also indicates why the initial Gibbs-ensemble studies on the Lennard-Jones fluids did not show significant finite-size effects. All these studies used Eqs. (6.6.12) and (6.6.13) to determine the critical point.

If we use these equations we implicitly assume nonclassical behavior up to the critical point. In Fig. L.5, this corresponds to extrapolating the lines, fitted to the data point, for $\beta = 1/8$. Extrapolation of these lines to $M_L^8(T) = 0$ gives a critical point that is not only independent of this system size but also very close to the true critical point of the infinite system.

For the three-dimensional Lennard-Jones fluid Panagiotopoulos did not observe a crossover from Ising-like to mean field behavior in the temperature regime that could be studied conveniently in the Gibbs ensemble ($T < 0.98T_c$). Also for liquid-liquid equilibria for the square well fluid, Recht and Panagiotopoulos [735] and de Miguel et al. [246] did not observe such a crossover. Moreover, for the three-dimensional Lennard-Jones fluid, the finite-size effects were negligible away from T_c and very small close to T_c .

^a The Ising model is equivalent to a lattice-gas model of a fluid. The latter model is the simplest that exhibits a liquid-vapor transition.

L.4.3 Chemical potential in the Gibbs ensemble

One of the steps in the Gibbs ensemble involves the insertion of a particle in one of the boxes. During this step, the energy of this particle has to be calculated (see section 6.6.3). Since this energy corresponds to the energy of a test particle, we can use the Widom insertion method [736] to calculate the chemical potential without additional costs [213]. At this point it is important to note that the Gibbs method requires no computation of the chemical potentials. However, to test whether the system under consideration has reached equilibrium or for comparison with other results, it is important to calculate the chemical potential of the individual phases correctly. The original Widom expression is valid only in the N, V, T ensemble and can be modified for applications in other ensembles (see section 8.5.1). Here we derive an expression for the chemical potential for the Gibbs ensemble. We restrict ourselves to temperatures sufficiently far below the critical temperature that the two boxes, after equilibration, do not change identity. For the more general case, we refer to [217].

If we rescale the coordinates of the particles with the box length, the partition function for the Gibbs ensemble (6.6.1) becomes

$$\begin{aligned}
 \bar{Q}_{N,V,T} &\equiv \frac{1}{V \Lambda^{3N} N!} \sum_{n_1=0}^N \binom{N}{n_1} \int_0^V dV_1 V_1^{n_1} (V - V_1)^{N-n_1} \\
 &\quad \times \int ds_1^{n_1} \exp[-\beta \mathcal{U}_1(n_1)] \int ds_2^{N-n_1} \exp[-\beta \mathcal{U}_2(N-n_1)] \\
 &= \sum_{n_1=0}^N \int_0^V dV_1 V_1^{n_1} (V - V_1)^{N-n_1} Q_1(n_1, V_1) Q_2(N-n_1, V-V_1),
 \end{aligned}
 \tag{L.4.25}$$

where $\mathbf{s} = \mathbf{r}/L$ is the scaled coordinates of a particle, L is the box length of the subsystem in which the particle is located, and $Q_i(n_i, V_i)$ is the partition of the canonical ensemble (see also section L.4.1.2).

The chemical potential of box 1 can be defined as

$$\begin{aligned} \mu_1 \equiv & -k_B T \ln \sum_{n_1=0}^N \int_0^V dV_1 V_1^{n_1} (V - V_1)^{N-n_1} \\ & \times \left[\frac{Q_1(n_1 + 1, V_1)}{Q_1(n_1, V_1)} \right] Q_2(N - n_1, V - V_1). \end{aligned} \quad (\text{L.4.26})$$

For the ratio of the partition functions of box 1, we can write

$$\begin{aligned} \frac{Q_1(n_1 + 1, V_1)}{Q_1(n_1, V_1)} &= \frac{V_1}{(n_1 + 1)\Lambda^3} \frac{\int d\mathbf{s}_1^{n_1+1} \exp[-\beta\mathcal{U}_1(n_1 + 1)]}{\int d\mathbf{s}_1^{n_1} \exp[-\beta\mathcal{U}_1(n_1)]} \\ &= \frac{V_1}{(n_1 + 1)\Lambda^3} \frac{\int d\mathbf{s}_1^{n_1} \exp[-\beta\Delta\mathcal{U}_1^+] \exp[-\beta\mathcal{U}_1(n_1)]}{\int d\mathbf{s}_1^{n_1} \exp[-\beta\mathcal{U}_1(n_1)]}, \end{aligned} \quad (\text{L.4.27})$$

in which we have used the notation

$$\mathcal{U}_1(n_1 + 1) = \Delta\mathcal{U}_1^+ + \mathcal{U}_1(n_1),$$

where $\Delta\mathcal{U}_1^+$ is the test particle energy of a (ghost) particle in box 1. We can write Eq. (L.4.26) as an ensemble average restricted to box 1:

$$\mu_1 = -k_B T \ln \frac{1}{\Lambda^3} \left\langle \frac{V_1}{n_1 + 1} \exp[-\beta\Delta\mathcal{U}_1^+] \right\rangle_{\text{Gibbs, box 1}}, \quad (\text{L.4.28})$$

where $\langle \cdots \rangle_{\text{Gibbs, box } i}$ denotes an ensemble average in the Gibbs ensemble restricted to box i (note that this ensemble average is well defined if the boxes do not change identity during a simulation [217]).

L.4.4 Algorithms of the Gibbs ensemble

L.5 Multi-canonical ensemble method

In the multi-canonical ensemble method, we extend our ensemble by using the energy space [350]. In the previous sections, we used different tricks to obtain information on parts of phase space that are rarely sampled in conventional simulations. The idea of a multi-canonical ensemble is to ensure that all energies are sampled uniformly. The probability of finding a system with a particular energy is given by

$$\mathcal{P}(U) = \mathcal{N}(U)w(U),$$

Algorithm 42 (Attempt to change the volume in the Gibbs ensemble)

function mcvol	trial volume change of box 1/2 at constant total volume
vol=box1**3	volume box 1 (diameter box1)
vo2=v-vol	...and 2
eno1= toterg (vol,1)	energy old conf. box 1
eno2= toterg (vo2,2)	and 2
lnvn=log(vol/vo12)+	random walk in $\ln V_1/V_2$
+ ($\mathcal{R}-0.5$)*vmax	
vln=v*exp(lnvn)/(1+exp(lnvn))	new volume box 1 and 2
v2n=v-vln	
box1n=vln**(1/3)	new box length box 1
box2n=v2n**(1/3)	new box length box 2
for 1 ≤ i ≤ npart do	
if ibox(i) == 1 then	determine which box
fact=box1n/box1o	
else	
fact=box2n/box2o	
endif	
x(i)=x(i)*fact	rescale positions
enddo	
en1n= toterg (vln,1)	total energy box 1
en2n= toterg (v2n,2)	total energy box 2
arg1=-beta*(en1n-eno1)	
+(npbox(1)+1)*log(v1n/v1o)	appropriate weight function
arg2=-beta*(en2n-eno2)	acceptance rule (6.6.10)
+(npbox(2)+1)*log(v2n/v2o)	
if $\mathcal{R} > \exp(\arg1+\arg2)$ then	
do i=1,npart	REJECTED
if ibox(i)== 1 then	determine which box
fact=box1o/box1n	
else	
fact=box2o/box2n	
endif	
x(i)=x(i)*fact	restore old configuration
enddo	
endif	
end function	

where $\mathcal{N}(U)$ is the density of states and $w(U)$ the particular weight in the ensemble. If we use $w(U) = \mathcal{C}/\mathcal{N}(U)$, we have a flat distribution of energies.

In practice, we do not know the density of states so we need to find this iteratively. For example, if we start with an ordinary N, V, T simulation, and

Specific Comments (for general comments, see p. 7)

1. The term `ibox(i) = 1` indicates that particle i is in box 1; `npart = npbox(1) + npbox(2)` where `npbox(i)` gives the number of particles in box i .
2. In this algorithm we perform a random walk in $\ln V$ and we use acceptance rule (6.6.10).
3. The function **toterg** is not shown explicitly: it is similar to Algorithm 5 and calculates the total energy of one of the two boxes. It requires information about the identity of the box (1 or 2) and the volume of these boxes (here assumed to be cubic).
In most cases the energy of the old configuration is known, and therefore it is not necessary to determine this energy at the beginning of the volume step.

we can make a histogram of the occurrence of a particular value of the energy $H_j(U)$, we can use this to improve our bias potential [737]:

$$\mathcal{U}_{j+1}^{\text{bias}}(U) = \mathcal{U}_j^{\text{bias}}(U) + \frac{1}{\beta} [\ln \mathcal{H}_j(U) - \ln \bar{\mathcal{H}}_j]$$

where $\bar{\mathcal{H}}_j$ is the average value of histogram. Those energies that occur more than the average get an unfavorable bias, a higher energy, and those that occur lower than the average a favorable bias. If the histogram is flat, the bias potential does not change in the next iteration. To see how this can be used to compute a free-energy (difference) in the canonical ensemble we have:

$$\begin{aligned} \mathcal{P}(U) &= \frac{\int d\Gamma^N \delta(U(\Gamma^N) - U) \exp(-\beta U(\Gamma^N))}{\int d\Gamma^N \exp(-\beta U(\Gamma^N))} \\ &= \exp(-\beta U) \frac{\int d\Gamma^N \delta(U(\Gamma^N) - U)}{\mathcal{C}_0} \end{aligned}$$

and in the multi-canonical ensemble, we have

$$\begin{aligned} \mathcal{P}^{\text{MulCan}}(U) &= \frac{\int d\Gamma^N \delta(U(\Gamma^N) - U) \exp[-\beta (U(\Gamma^N) + \mathcal{U}^{\text{bias}}(U(\Gamma^N)))]}{\int d\Gamma^N \exp[-\beta (U(\Gamma^N) + \mathcal{U}^{\text{bias}}(U(\Gamma^N)))]} \\ &= \exp[-\beta (U + \mathcal{U}^{\text{bias}}(U))] \frac{\int d\Gamma^N \delta(U(\Gamma^N) - U)}{\mathcal{C}^{\text{MulCan}}} \end{aligned}$$

Or, for the ratio of these two probabilities:

$$\frac{\mathcal{P}(U)}{\mathcal{P}^{\text{MulCan}}(U)} = C \frac{\exp(-\beta U)}{\exp[-\beta (U + \mathcal{U}^{\text{bias}}(U))]} = C \exp[\beta \mathcal{U}^{\text{bias}}(U)].$$

Algorithm 43 (Attempt to swap a particle between the two boxes)

<pre> function mcswap if $\mathcal{R} < 0.5$ then in=1 out=2 else in=2 out=1 endif xn=$\mathcal{R}$*box(in) enn = ener(xn, in) w(in)=w(in)+vol(in)* + exp(-beta*enn)/(npbox(in)+1) if (npbox(out) == 0) return ido=0 while ido $\neq$ out do o=int(npart*$\mathcal{R}$)+1 ido=ibox(o) end while eno= ener(x(o), out) arg=exp(-beta*(enn-eno + + log(vol(out)*(npbox(in)+1)/ + (vol(in)*npbox(out)))/beta)) if $\mathcal{R} < \arg$ then x(o)=xn ibox(o)=in nbox(out)=npbox(out)-1 nbox(in)=npbox(in)+1 endif end function </pre>	attempts to swap a particle between the two boxes which box to add or remove new particle at a random position energy new particle in box in update chemical potential (L.4.28) if box empty return find a particle to be removed energy particle o in box out acceptance rule (6.6.11) add new particle to box in
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Specific Comments (for general comments, see p. 7)

1. **ener**(x, ib) calculates the energy of a particle at position x in box ib . It carries a box label and is slightly different from the function called in Algorithm 2.
2. We specify an additional argument for the function **ener**: the index of the box where we attempt to insert (in) or remove (out) a particle.
3. The acceptance rule (6.6.11) is used in this algorithm.
4. We also compute the Boltzmann factor associated with the random insertion of a particle, at virtually no added cost. At the end of the simulation, the excess chemical potential can be calculated from $w(\text{box})$ using $\mu_{\text{box}} = -\ln \langle w_{\text{box}} \rangle / \beta$, where $\langle w_{\text{box}} \rangle$ is the average of w_{box} accumulated during the run.

If we achieve an exactly flat distribution, $\mathcal{P}^{\text{MulCan}}(U) = C$, we have for the free energy:

$$-\beta F(U) = \ln \mathcal{P}(U) + C = \beta \mathcal{U}^{\text{bias}}(U).$$

This illustrates that in order to get a perfectly flat distribution, we need to apply a biasing potential that is the inverse of the free energy. In practice, we will never get a perfectly flat distribution, and the biasing potential becomes equivalent to the umbrella sampling technique, which we discuss in section 8.6.6.

L.6 Nosé-Hoover dynamics

L.6.1 Nosé-Hoover dynamics equations of motion

We now apply the methods of non-Hamiltonian dynamics to analyze the Nosé-Hoover algorithm and the Nosé-Hoover chains that are discussed in section 7.1.2.1 and Appendix L.6.1.2. Our discussion of these algorithms is only intended as an illustration that there exist systematic techniques for predicting the phase-space density generated by a particular non-Hamiltonian dynamics scheme. While we show a few simple examples, we refer the reader to ref. [267] for a more comprehensive discussion.

L.6.1.1 The Nosé-Hoover algorithm

In section 7.1.2.1 we showed that the Nosé-Hoover algorithm generates non-Hamiltonian dynamics. The Nosé-Hoover equations can be written as

$$\begin{aligned}\dot{\mathbf{r}}_i &= \mathbf{p}_i / m_i \\ \dot{\mathbf{p}}_i &= \mathbf{F}_i - \frac{p_\xi}{Q} \mathbf{p}_i \\ \dot{\xi} &= p_\xi / Q \\ \dot{p}_\xi &= \sum_i \mathbf{p}_i^2 / m_i - k_B T L,\end{aligned}$$

where L is a parameter that has to be determined to generate the canonical distribution.

Implementation

In section 7.1.2.1, we showed how the introduction of an additional dynamical variable (s) in the Lagrangian could be used to perform MD simulations subject to a thermodynamic constraint (in this case, constant temperature). We stress that the importance of such extended-Lagrangian techniques transcends the specific application. In addition, the problems encountered in the numerical implementation of the Nosé scheme are representative of a wider class of algorithms (namely, those where forces depend explicitly on velocities). It is for this

reason that we discuss the numerical implementation of the Nosé thermostat in some detail (see also SI L.6.2).

The Nosé equations of motion can be written in terms of virtual variables or real variables. In a simulation based on the virtual variables, the real-time step is fluctuating, which complicates the sampling of averages. For this reason, it is better to use the real-variable formulation.

It is now common to use the Nosé scheme in the formulation of Hoover [257,738], who showed that the equations derived by Nosé could be written in a form that resembles Newtonian dynamics.

In Eqs. (7.1.27), (7.1.28), and (7.1.29), the variables s' , p'_s , and Q occur only as $s'p'_s/Q$. To simplify these equations, we can introduce the thermodynamic friction coefficient $\xi = s'p'_s/Q$. The equations of motion then become (dropping the primes on $\mathbf{p}$) and using dots to denote time derivatives)

$$\dot{\mathbf{r}}_i = \mathbf{p}_i / m_i \quad (\text{L.6.1})$$

$$\dot{\mathbf{p}}_i = -\frac{\partial \mathcal{U}(\mathbf{r}^N)}{\partial \mathbf{r}_i} - \xi \mathbf{p}_i \quad (\text{L.6.2})$$

$$\dot{\xi} = \left(\sum_i \mathbf{p}_i^2 / m_i - \frac{L}{\beta} \right) / Q \quad (\text{L.6.3})$$

$$\dot{s}/s = \frac{d \ln s}{dt} = \xi. \quad (\text{L.6.4})$$

Note that the last equation is redundant, since Eqs. (L.6.1)–(L.6.3) form a closed set. However, if we solve the equations of motion for s as well, we can use Eq. (7.1.30) as a diagnostic tool, since H' must be conserved during the simulation. In terms of the variables used in Eqs. (L.6.1)–(L.6.4), H reads

$$H_{\text{Nosé}} = \sum_{i=1}^N \frac{\mathbf{p}_i^2}{2m_i} + \mathcal{U}(\mathbf{r}^N) + \frac{\xi^2 Q}{2} + L \frac{\ln s}{\beta}. \quad (\text{L.6.5})$$

As we use the real-variable formulation in this set of equations, we have to take $L = dN$.

An important implication of the Nosé equations is that in the Lagrangian (7.1.10), a logarithmic term ($\ln s$) is needed to have the correct scaling of time. Any scheme that does not rescale time will fail to recover the canonical ensemble.

Hoover [738] demonstrated that the equations of motion (L.6.1)–(L.6.3) are unique, in the sense that other equations of the same form cannot lead to a canonical distribution. In SI L.6.2, we discuss an efficient way of implementing the Nosé-Hoover scheme.

The equations of motion of the Nosé-Hoover scheme cannot be derived from a Hamiltonian. This implies that one cannot use the standard methods (see Appendix A) to make the time averages obtained with this dynamics map onto

ensemble averages of the standard form. In Appendix B, we discuss how one can analyze such non-Hamiltonian dynamics. The result of this analysis is that the conventional Nosé-Hoover algorithm only generates the correct distribution if there is a *single* constant of motion. Normally, the total energy defined by H_{Nose} , see Eq. (L.6.5), is conserved. But the existence of other conserved quantities creates a problem. This implies that one should not have any other conserved quantity, such as the total momentum. Momentum conservation is not an issue in simulations of systems confined by walls, or subject to another external potential that breaks momentum conservation.

However, if we simulate a system without external forces, momentum is conserved. Under those conditions, the Nosé-Hoover scheme can still be correct, provided that the center of mass remains fixed. This condition can be fulfilled if we set the initial velocity of the center of mass to zero. However, if we simulate systems with momentum conservation and a non-zero velocity of the center of mass, or if we have additional conserved quantities, then we must go beyond the simple NH algorithm and use chains of Nosé-Hoover thermostats [267,269] to obtain the correct canonical distribution.

L.6.1.2 Nosé-Hoover chains

The equations of motion for a system of N particles coupled with M Nosé-Hoover chains are given (in real variables, hence $L = dN$) by

$$\dot{\mathbf{r}}_i = \frac{\mathbf{p}_i}{m_i} \quad (\text{L.6.6})$$

$$\dot{\mathbf{p}}_i = \mathbf{F}_i - \frac{p_{\xi_1}}{Q_1} \mathbf{p}_i \quad (\text{L.6.7})$$

$$\dot{\xi}_k = \frac{p_{\xi_k}}{Q_k} \quad k = 1, \dots, M \quad (\text{L.6.8})$$

$$\dot{p}_{\xi_1} = \left(\sum_i \frac{p_i^2}{m_i} - Lk_B T \right) - \frac{p_{\xi_2}}{Q_2} p_{\xi_1} \quad (\text{L.6.9})$$

$$\dot{p}_{\xi_k} = \left[\frac{p_{\xi_{k-1}}^2}{Q_{k-1}} - k_B T \right] - \frac{p_{\xi_{k+1}}}{Q_{k+1}} p_{\xi_k} \quad (\text{L.6.10})$$

$$\dot{p}_{\xi_M} = \left[\frac{p_{\xi_{M-1}}^2}{Q_{M-1}} - k_B T \right]. \quad (\text{L.6.11})$$

For these equations of motion, the conserved energy is

$$H_{\text{NHC}} = \mathcal{H}(\mathbf{r}, \mathbf{p}) + \sum_{k=1}^M \frac{p_{\xi_k}^2}{2Q_k} + Lk_B T \xi_1 + \sum_{k=2}^M k_B T \xi_k. \quad (\text{L.6.12})$$

We can use this conserved quantity to check the integration scheme. It is important to note that the additional $M - 1$ equations of motion form a simple one-

dimensional chain and therefore are relatively simple to implement. In SI L.6.2, we describe an algorithm for a system with a Nosé-Hoover chain thermostat.

Example 31 (Nosé-Hoover chain for harmonic oscillator). The harmonic oscillator is the obvious model system on which we test the Nosé-Hoover chain thermostat. If we use a chain of two coupling parameters, the equations of motion are

$$\begin{aligned}\dot{r} &= v \\ \dot{v} &= -r - \xi_1 v \\ \dot{\xi}_1 &= \frac{v^2 - T}{Q_1} - \xi_1 \xi_2 \\ \dot{\xi}_2 &= \frac{Q_1 \xi_1^2 - T}{Q_2}.\end{aligned}$$

A typical trajectory generated with the Nosé-Hoover chains is shown in Fig. L.6. The distribution of the velocity and position of the oscillator are also shown in Fig. L.6. Comparison with the results obtained using the Andersen thermostat (see Case Study 13) shows that the Nosé-Hoover chains do generate a canonical distribution, even for the harmonic oscillator.

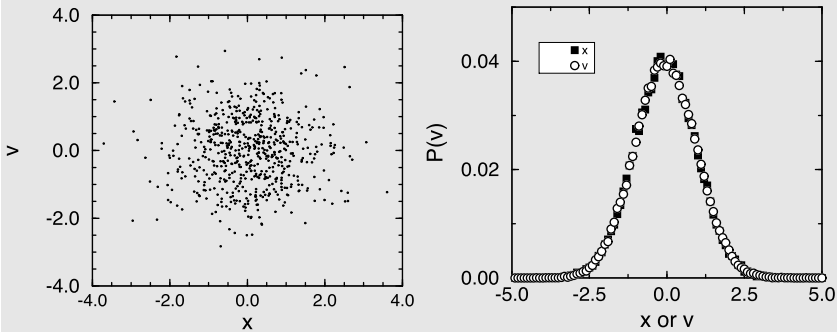


FIGURE L.6 Test of the phase space trajectory of a harmonic oscillator, coupled to a Nosé-Hoover chain thermostat. The left-hand side of the figure shows part of a trajectory: the dots correspond to consecutive points separated by 10,000 time steps. The right-hand side shows the distributions of velocity and position. Due to our choice of units, both distributions should be Gaussians of equal width.

The Fortran code to generate this Example can be found in the online-SI, Case Study 14.

To analyze the dynamics of this system, we have to determine the conservation laws and the non-driven variables. Let us consider first the case in which

we assume that we only have conservation of energy, viz. Eq. (7.1.13)

$$\begin{aligned} H_{\text{Nose}} &= \sum_{i=1}^N \frac{\mathbf{p}_i^2}{2m_i} + \mathcal{U}(\mathbf{r}^N) + \frac{p_\xi^2}{2Q} + Lk_B T \xi \\ &= \mathcal{H}(\mathbf{r}, \mathbf{p}) + \frac{p_\xi^2}{2Q} + Lk_B T \xi = C_1, \end{aligned}$$

where $\mathcal{H}(\mathbf{r}, \mathbf{p})$ is the physical Hamiltonian. If we use $\mathbf{\Gamma} = (\mathbf{r}^N, \mathbf{p}^N, \xi, p_\xi)$, the phase-space compressibility of this system can be written as

$$\begin{aligned} \kappa(\mathbf{\Gamma}) &= \nabla_{\mathbf{\Gamma}} \cdot \dot{\mathbf{\Gamma}} \\ &= \sum_i \nabla_{\mathbf{r}_i} \cdot \dot{\mathbf{r}}_i + \sum_i \nabla_{\mathbf{p}_i} \cdot \dot{\mathbf{p}}_i + \nabla_{\xi} \cdot \dot{\xi} + \nabla_{p_\xi} \cdot \dot{p}_\xi \\ &= \sum_i \nabla_{\mathbf{p}_i} \cdot \dot{\mathbf{p}}_i \\ &= -dN p_\xi / Q = -dN \dot{\xi}. \end{aligned}$$

Hence, it follows that the metric $\sqrt{g}$ is given by

$$\sqrt{g} = \exp \left[- \int \kappa dt \right] = \exp(dN \xi).$$

Substitution of this metric in the expression for the partition function gives:

$$\begin{aligned} \Omega_T(N, V, C_1) &= \int d^N \mathbf{p} \int d^N \mathbf{r} \int dp_\xi \int d\xi \\ &\quad \times \exp(dN \xi) \delta \left(\mathcal{H}(\mathbf{r}, \mathbf{p}) + \frac{p_\xi^2}{2Q} + Lk_B T \xi - C_1 \right). \end{aligned}$$

In this expression, the integration over ξ and p_ξ can be performed analytically. Because of the δ -function, integration over ξ gives as condition

$$\xi = \frac{1}{Lk_B T} \left(C_1 - \mathcal{H}(\mathbf{r}, \mathbf{p}) - \frac{p_\xi^2}{2Q} \right).$$

Substitution of this condition in the partition function gives

$$\begin{aligned} \Omega_T(N, V, C_1) &= \frac{1}{Lk_B T} \int d^N \mathbf{p} \int d^N \mathbf{r} \int dp_\xi \\ &\quad \times \exp \left[\frac{dN}{Lk_B T} \left(C_1 - \mathcal{H}(\mathbf{r}, \mathbf{p}) - \frac{p_\xi^2}{2Q} \right) \right] \\ &= \frac{\exp(dN C_1 / Lk_B T)}{Lk_B T} \int dp_\xi \exp \left(-dN p_\xi^2 / 2QL \right) \end{aligned}$$

$$\begin{aligned} & \times \int d^N \mathbf{p} \int d^N \mathbf{r} \exp(-\beta d N \mathcal{H}(\mathbf{r}, \mathbf{p})/L) \\ & \propto Q(N, V, T), \end{aligned}$$

where the last equality only holds provided that we choose L equal to dN . The integration over p_ξ yields a constant prefactor that has no physical importance. In section 7.1.2.1, we derived a similar result for the Nosé equations in terms of its real variables. The present demonstration that the Nosé-Hoover equations lead to a canonical distribution is completely different from Nosé's original argument, yet the end result is the same. Note, however, that we have assumed that there is only a single conservation law, viz. conservation of $H_{\text{Nosé}}$. In general, there will be more conserved quantities. For instance, if we consider a system in the absence of external forces, then the total linear momentum is also conserved. This will affect the phase-space distribution. In the Nosé-Hoover dynamics, conservation of total momentum reads

$$\begin{aligned} \frac{d\mathbf{P}e^\xi}{dt} &= e^\xi (\dot{\mathbf{P}} + \mathbf{P}\dot{\xi}) = e^\xi \left(\dot{\mathbf{P}} + \mathbf{P} \frac{p_\xi}{Q} \right) \\ &= e^\xi \left(\sum_i \left(\mathbf{F}_i - \frac{p_\xi}{Q} \mathbf{p}_i \right) + \mathbf{P} \frac{p_\xi}{Q} \right) \\ &= 0, \end{aligned}$$

and hence

$$\mathbf{P}e^\xi = \mathbf{K}, \quad (\text{L.6.13})$$

where $\mathbf{P} = \sum_i \mathbf{p}_i$ is the center-of-mass momentum of the system and $\mathbf{K}$ is an arbitrary constant vector.

To continue the analysis, we should eliminate the driven variables from our system. The center-of-mass position, $\mathbf{R}$, and momentum, $\mathbf{P}$, have no influence on the other variables. We can eliminate these by considering the positions and momenta relative to the center of mass of the system, $\mathbf{r}'$ and $\mathbf{p}'$, respectively. Note, however, that the magnitude of the center-of-mass momentum is coupled to the other variables through a conservation law and cannot be eliminated from the analysis. The components of the center-of-mass momentum $\mathbf{P}$ are linearly dependent.¹ Therefore of the d components, only one component can be chosen independently, or we can take as independent variable $P = (\sum_\alpha P_\alpha^2)^{1/2}$.

¹ To see this, consider the components of Eq. (L.6.13):

$$\frac{P_x}{K_x} = \frac{P_y}{K_y} = \frac{P_z}{K_z} = e^\xi,$$

which shows that only one of the components is independent.

We now have to perform a transformation of our systems to the variables $\{\mathbf{p}', \mathbf{P}, \mathbf{r}', \mathbf{R}\}$; the resulting equations of motion are

$$\begin{aligned}\dot{\mathbf{r}}'_i &= \mathbf{p}'_i / m'_i \\ \dot{\mathbf{p}}'_i &= \mathbf{F}'_i - \frac{p_\xi}{Q} \mathbf{p}'_i \\ \dot{P} &= -\frac{p_\xi}{Q} P_i \\ \dot{\xi} &= \frac{p_\xi}{Q} \\ \dot{p}_\xi &= \sum_i^{N-1} \frac{\mathbf{p}'_i{}^2}{m'_i} + \frac{P^2}{M} - k_B T L.\end{aligned}$$

The equations of motion have two² conservation laws:

$$\begin{aligned}\mathcal{H}(\mathbf{p}', \mathbf{r}', P) + \frac{p_\xi^2}{2Q} + L K_B T \xi &= C_1 \\ P \exp(\xi) &= C_2.\end{aligned}$$

In the first conservation law, we have used

$$\mathcal{H}(\mathbf{r}', \mathbf{p}', P) = \sum_{i=1}^{N-1} \frac{\mathbf{p}'_i{}^2}{2m'_i} + \frac{P^2}{2M} + \mathcal{U}(\mathbf{r}'^N) = \mathcal{H}(\mathbf{r}, \mathbf{p}).$$

To compute the partition function, we have to determine the metric from the compressibility:

$$\begin{aligned}\kappa &= \sum_i^{N-1} \nabla_{\mathbf{r}'_i} \cdot \dot{\mathbf{r}}'_i + \sum_i^{N-1} \nabla_{\mathbf{p}'_i} \cdot \dot{\mathbf{p}}'_i + \nabla_P \cdot \dot{P} + \nabla_\xi \cdot \dot{\xi} + \nabla_{p_\xi} \cdot \dot{p}_\xi \\ &= -[d(N-1) + 1] \dot{\xi}.\end{aligned}$$

From which the metric follows directly:

$$\sqrt{g} = \exp \{ [d(N-1) + 1] \xi \}.$$

The partition function Ω contains two δ -functions that express the two conservation laws:

$$\Omega_T(N, V, C_1, C_2) = \int d^{N-1} \mathbf{p}' \int d^{N-1} \mathbf{r}' \int dP \int dp_\xi \int d\xi$$

² Because we have replaced the d center-of-mass momenta components by a single variable P only one conservation law for the momenta is left.

$$\times \exp \{[d(N-1)+1]\xi\} \delta \left(\mathcal{H}(\mathbf{r}', \mathbf{p}', P) + \frac{p_\xi^2}{2Q} + Lk_B T \xi - C_1 \right) \\ \delta (e^\xi P - C_2).$$

The second δ -function imposes that

$$\xi = \ln(C_2/P).$$

Hence, integrating over ξ yields

$$\Omega_T(N, V, C_1, C_2) = \frac{1}{C_2} \int d^{N-1} \mathbf{p}' \int d^{N-1} \mathbf{r}' \int dP \int dp_\xi \\ \times \left(\frac{C_2}{P} \right)^{d(N-1)+1} \delta \left(\mathcal{H}(\mathbf{r}', \mathbf{p}', P) + \frac{p_\xi^2}{2Q} + Lk_B T \ln(C_2/P) - C_1 \right).$$

The remaining δ -function fixes p_ξ :

$$p_\xi = \{2Q [C_1 - \mathcal{H}(\mathbf{r}', \mathbf{p}', P) - Lk_B T \ln(C_2/P)]\}^{-\frac{1}{2}}.$$

Integration over p_ξ then results in the following expression for Ω :

$$\Omega_T(N, V, C_1, C_2) = \frac{\sqrt{2Q}}{C_2} \int d^{N-1} \mathbf{p}' \int d^{N-1} \mathbf{r}' \\ \times \int dP \left(\frac{C_2}{P} \right)^{d(N-1)+1} [C_1 - \mathcal{H}(\mathbf{r}', \mathbf{p}', P) - Lk_B T \ln(C_2/P)]^{-\frac{1}{2}}.$$

Note that this is *not* the partition function for a canonical ensemble. This problem was first pointed out by Cho et al. [739]. Only in the case that $C_2 = 0$ can the conventional Nosé-Hoover scheme generate a canonical distribution. If $C_2 = 0$ the partition function reads

$$\Omega_T(N, V, C_1, 0) = \int d^{N-1} \mathbf{p}' \int d^{N-1} \mathbf{r}' \int dP \int dp_\xi \int d\xi \\ \times \exp \{[d(N-1)+1]\xi\} \delta \left(\mathcal{H}(\mathbf{r}', \mathbf{p}', P) + \frac{p_\xi^2}{2Q} + Lk_B T \xi - C_1 \right) \\ \times \delta (e^\xi P - 0).$$

The δ -function imposes that $P = 0$. Integration over P then yields

$$\Omega_T(N, V, C_1, 0) = \int d^{N-1} \mathbf{p}' \int d^{N-1} \mathbf{r}' \int dp_\xi \int d\xi \\ \times \exp \{[d(N-1)+1]\xi\} \exp(-\xi)$$

$$\times \delta \left(\mathcal{H}(\mathbf{r}', \mathbf{p}') + \frac{p_\xi^2}{2Q} + Lk_B T \xi - C_1 \right).$$

The other δ -function fixes ξ

$$\xi = -\frac{\beta}{L} \left[\mathcal{H}(\mathbf{r}', \mathbf{p}') + \frac{p_\xi^2}{2Q} - C_1 \right],$$

and we finally obtain

$$\begin{aligned} \Omega_T(N, V, C_1, 0) &= \frac{\exp[\beta d(N-1)C_1/L]}{Lk_B T} \\ &\times \int d p_\xi \exp \left[-\beta d(N-1) p_\xi^2 / (2QL) \right] \\ &\times \int d^{N-1} \mathbf{p}' \int d^{N-1} \mathbf{r}' \exp \left[-\beta \frac{d(N-1)}{L} \mathcal{H}(\mathbf{r}', \mathbf{p}') \right] \\ &\propto \int d^{N-1} \mathbf{p}' \int d^{N-1} \mathbf{r}' \exp \left[-\beta \frac{d(N-1)}{L} \mathcal{H}(\mathbf{r}', \mathbf{p}') \right]. \end{aligned}$$

Clearly, if we choose $L = d(N-1)$, then the correct canonical partition function is recovered. In practice, most conventional Nosé-Hoover simulations are performed with a fixed center of mass and therefore obey the condition $P = 0$.

L.6.1.3 The NPT ensemble

For the N, P, T ensemble the equations of motion are (see also section 7.2)

$$\begin{aligned} \dot{\mathbf{r}}_i &= \frac{\mathbf{p}_i}{m_i} + \frac{p_\epsilon}{W} \mathbf{r}_i \\ \dot{\mathbf{p}}_i &= \mathbf{F}_i - \left(1 + \frac{1}{N} \right) \frac{p_\epsilon}{W} \mathbf{p}_i - \frac{p_{\xi_1}}{Q_1} \mathbf{p}_i \\ \dot{V} &= dV \frac{p_\epsilon}{W} \\ \dot{p}_\epsilon &= dV (P_{\text{int}} - P_{\text{ext}}) + \frac{1}{N} \sum_{i=1}^N \frac{\mathbf{p}_i^2}{m_i} - \frac{p_{\xi_1}}{Q_1} p_\epsilon \\ \dot{\xi}_k &= \frac{p_{\xi_k}}{Q_k} \quad \text{for } k = 1, \dots, M \\ \dot{p}_{\xi_1} &= \sum_{i=1}^N \frac{\mathbf{p}_i^2}{m_i} + \frac{p_\epsilon^2}{W} - (dN+1)k_B T - \frac{p_{\xi_2}}{Q_2} p_{\xi_1} \\ \dot{p}_{\xi_k} &= \frac{p_{\xi_{k-1}}^2}{Q_{k-1}} - k_B T - \frac{p_{\xi_{k+1}}}{Q_{k+1}} p_{\xi_k} \quad \text{for } k = 2, \dots, M-1 \end{aligned}$$

$$\dot{p}_{\xi_M} = \frac{p_{\xi_{M-1}}^2}{Q_{M-1}} - k_B T.$$

To analyze the dynamics of this system, we have to consider two cases. In the case in which the sum of the forces is zero, $\sum_i \mathbf{F}_i = \mathbf{0}$, implies that we have additional conservation laws. The second case, $\sum_i \mathbf{F}_i \neq \mathbf{0}$, has only one conserved quantity; conservation of “energy”:

$$\begin{aligned} H' - C_1 &= \mathcal{H}(\mathbf{p}, \mathbf{r}) + \frac{p_\epsilon^2}{W} + \sum_{k=1}^M \frac{p_{\xi_k}^2}{Q_k} + (dN + 1)k_B T \xi_1 + k_B T \xi_c + P_{\text{ext}} V - C_1 \\ &= 0, \end{aligned}$$

where ξ_c is defined as the center of the thermostat

$$\xi_c = \sum_{k=2}^M \xi_k.$$

We first consider the case $\sum_i \mathbf{F}_i \neq \mathbf{0}$. To analyze its dynamics we have to compute the compressibility. The independent variables are³ $\Gamma = \mathbf{p}^N, \mathbf{r}^N, \xi_1, \xi_c, p_{\xi_1}, p_{\xi_c}, V, p_\epsilon$:

$$\kappa = \nabla \cdot \Gamma = -(dN + 1) \frac{p_{\xi_1}}{Q_1} - \frac{p_{\xi_c}}{Q_c},$$

which gives a phase-space metric

$$\sqrt{g} = \exp[(dN + 1)\xi_1 + \xi_c].$$

We can now write for the partition function

$$\begin{aligned} \Omega_{T, P_{\text{ext}}}(N, C_1) &= \int dV \int d^N \mathbf{p} \int d^N \mathbf{r} \int dp_{\xi_1} \int d\xi_1 \\ &\quad \times \int dp_{\xi_c} \int d\xi_c \int dp_\epsilon \exp[(dN + 1)\xi_1 + \xi_c] \delta(H' - C_1). \end{aligned}$$

The delta function gives a condition for ξ_1

$$\xi_1 = \frac{1}{(dN + 1)k_B T} \left(C_1 - \mathcal{H}(\mathbf{p}, \mathbf{r}) - \frac{p_\epsilon^2}{W} - \sum_{k=1}^M \frac{p_{\xi_k}^2}{Q_k} - k_B T \xi_c - P_{\text{ext}} V \right).$$

Substitution of this expression into the partition function gives

$$\Omega_{T, P_{\text{ext}}}(N, C_1) = \frac{\exp(\beta C_1)}{(dN + 1)k_B T} \int dV \int d^N \mathbf{p} \int d^N \mathbf{r} \int dp_{\xi_1} \int d\xi_1$$

³ A Nosé-Hoover chain of length M has two independent variables, we use ξ_1 and ξ_c .

$$\begin{aligned}
& \times \int dp_{\xi_c} \int d\xi_c \int dp_\epsilon \\
& \times \exp \left[-\beta \left(\mathcal{H}(\mathbf{p}, \mathbf{r}) + \frac{p_\epsilon^2}{W} + \sum_{k=1}^M \frac{p_{\xi_k}^2}{Q_k} + P_{\text{ext}} V \right) \right] \\
& \propto \int dV \exp(-\beta P_{\text{ext}} V) \int d^N \mathbf{p} \int d^N \mathbf{r} \exp[-\beta \mathcal{H}(\mathbf{p}, \mathbf{r})].
\end{aligned}$$

The integration over ξ_c gives a constant, which can be infinite but has no physical importance. This demonstrates that the desired distribution is generated.

At this point, we would like to emphasize that the original Nosé-Hoover algorithm does not generate this distribution. The reason is that the metric for this algorithm generates an additional $1/V$ term in the partition function. With the algorithm of Martyna et al., this term is removed. This point is explained in detail in ref. [267].

For the case $\sum_i \mathbf{F}_i = \mathbf{0}$, we have as additional conservation laws for the total momentum $\mathbf{P}$

$$\mathbf{P} \exp[(1 + 1/N)\epsilon + \xi_1] = \mathbf{K}.$$

Similar to the N, V, T ensemble, the components of P are linearly dependent, and the center-of-mass coordinates have to be eliminated from the analysis. This results in a set of equations of motion in coordinates relative to the center of mass. The details of this proof can be found in ref. [267]. Similar to the N, V, T ensemble, if we use $\mathbf{K} = \mathbf{0}$, we generate an $(N - 1)PT$ ensemble.

L.6.2 Nosé-Hoover algorithms

As discussed in section L.6.1.1, it is advantageous to implement the Nosé thermostat using the formulation of Hoover, Eqs. (L.6.1)–(L.6.4). Since the velocity also appears on the right-hand side of Eq. (L.6.2), this scheme cannot be implemented directly into the velocity Verlet algorithm (see also section 4.3). To see this, consider a standard constant- N, V, E simulation, for which the velocity Verlet algorithm is of the form

$$\begin{aligned}
r(t + \Delta t) &= r(t) + v(t)\Delta t + f(t)\Delta t^2/(2m) \\
v(t + \Delta t) &= v(t) + \frac{f(t + \Delta t) + f(t)}{2m}\Delta t.
\end{aligned}$$

When we use this scheme for the Nosé-Hoover equations of motion, we obtain for the positions and velocities

$$r_i(t + \Delta t) = r_i(t) + v_i(t)\Delta t + [f_i(t)/m_i - \xi(t)v_i(t)]\Delta t^2/2 \quad (\text{L.6.14})$$

$$\begin{aligned}
v_i(t + \Delta t) &= v_i(t) + [f_i(t + \Delta t)/m_i - \xi(t + \Delta t)v_i(t + \Delta t) \\
&\quad + f_i(t)/m_i - \xi(t)v_i(t)]\Delta t/2. \quad (\text{L.6.15})
\end{aligned}$$

The first step of the velocity Verlet algorithm can be carried out without difficulty. In the second step, we first update the velocity, using the old “forces” to the intermediate value $v(t + \Delta t/2) \equiv v'$. And then we must use the new “forces” to update v' :

$$v_i(t + \Delta t) = v'_i + [f_i(t + \Delta t)/m_i - \xi(t + \Delta t)v_i(t + \Delta t)] \Delta t/2. \quad (\text{L.6.16})$$

In these equations $v_i(t + \Delta t)$ appears on the right- and left-hand sides; therefore, these equations cannot be integrated exactly.⁴ For this reason the Nosé-Hoover method is usually implemented using a predictor-corrector scheme or solved iteratively [623]. This has a disadvantage that the solution is no longer time reversible. Martyna et al. [126] have developed a set of explicit reversible integrators using the Liouville approach (see section 4.3.4) for this type of extended systems.

L.6.2.1 Canonical ensemble

For M chains, the Nosé-Hoover equations of motion are given by (see also section L.6.1.2)

$$\begin{aligned} \dot{\mathbf{r}}_i &= \mathbf{p}_i/m_i \\ \dot{\mathbf{p}}_i &= \mathbf{F}_i - \frac{p_{\xi_1}}{Q_1} \mathbf{p}_i \\ \dot{\xi}_k &= \frac{p_{\xi_k}}{Q_k} \quad k = 1, \dots, M \\ \dot{p}_{\xi_1} &= \left(\sum_i p_i^2/m_i - Lk_B T \right) - \frac{p_{\xi_2}}{Q_2} p_{\xi_1} \\ \dot{p}_{\xi_k} &= \left[\frac{p_{\xi_{k-1}}^2}{Q_{k-1}} - k_B T \right] - \frac{p_{\xi_{k+1}}}{Q_{k+1}} p_{\xi_k} \\ \dot{p}_{\xi_M} &= \left[\frac{p_{\xi_{M-1}}^2}{Q_{M-1}} - k_B T \right]. \end{aligned}$$

The Liouville operator, iL , for the equations of motion is defined as (see section 4.3.4)

$$iL \equiv \dot{\eta} \frac{\partial}{\partial \eta}$$

with $\eta = (\mathbf{r}^N, \mathbf{p}^N, \xi^M, p_{\xi}^M)$. Using the equations of motion, $\mathbf{p}_i = m_i \mathbf{v}_i$, and $p_{\xi_k} = Q_k v_{\xi_k}$, we obtain as Liouville operator for the Nosé-Hoover chains

$$iL_{\text{NHC}} = \sum_{i=1}^N \mathbf{v}_i \cdot \nabla_{\mathbf{r}_i} + \sum_{i=1}^N \left[\frac{\mathbf{F}_i(\mathbf{r}_i)}{m_i} \right] \cdot \nabla_{\mathbf{v}_i} - \sum_{i=1}^N v_{\xi_1} \mathbf{v}_i \cdot \nabla_{\mathbf{v}_i} + \sum_{k=1}^M v_{\xi_k} \frac{\partial}{\partial \xi_k}$$

⁴ For the harmonic oscillator, it is possible to find an analytic solution (see Case Study 13).

$$+ \sum_{k=1}^{M-1} (G_k - v_{\xi_k} v_{\xi_{k+1}}) \frac{\partial}{\partial v_{\xi_k}} + G_M \frac{\partial}{\partial v_{\xi_M}}$$

with

$$G_1 = \frac{1}{Q_1} \left(\sum_{i=1}^N m_i \mathbf{v}_i^2 - L k_B T \right)$$

$$G_k = \frac{1}{Q_k} \left(Q_{k-1} v_{\xi_{k-1}}^2 - k_B T \right).$$

As explained in section 4.3.4, the Liouville equation combined with the Trotter formula is a powerful technique for deriving a time-reversible algorithm for solving the equations of motion numerically. Here we will use this technique to derive such a scheme for the Nosé-Hoover thermostats. We use a simplified version; a more complete description can be found in ref. [126].

We have to make an intelligent separation of the Liouville operator. The first step is to separate the part of the Liouville operator that only involves the positions (iL_r) and the velocities (iL_v) from the parts that involve the Nosé-Hoover thermostats (iL_C):

$$iL_{\text{NHC}} = iL_r + iL_v + iL_C$$

with

$$iL_r = \sum_{i=1}^N \mathbf{v}_i \cdot \nabla_{\mathbf{r}_i}$$

$$iL_v = \sum_{i=1}^N \frac{\mathbf{F}_i(\mathbf{r}_i)}{m_i} \cdot \nabla_{\mathbf{v}_i}$$

$$iL_C = \sum_{k=1}^M v_{\xi_k} \frac{\partial}{\partial \xi_k} - \sum_{i=1}^N v_{\xi_1} \mathbf{v}_i \cdot \nabla_{\mathbf{v}_i}$$

$$+ \sum_{k=1}^{M-1} (G_k - v_{\xi_k} v_{\xi_{k+1}}) \frac{\partial}{\partial v_{\xi_k}} + G_M \frac{\partial}{\partial v_{\xi_M}}.$$

There are several ways to factorize iL_{NHC} using the Trotter formula; we follow the one used by Martyna et al. [126]:

$$e^{(iL\Delta t)} = e^{(iL_C \Delta t/2)} e^{(iL_v \Delta t/2)} e^{(iL_r \Delta t)} e^{(iL_v \Delta t/2)} e^{(iL_C \Delta t/2)} + \mathcal{O}(\Delta t^3). \quad (\text{L.6.17})$$

The Nosé-Hoover chain part L_C has to be further factorized. Here, we will do this for a chain of length $M = 2$; the more general case is discussed in ref. [126].

The Nosé-Hoover part of the Liouville operator for this chain length can be separated into five terms:

$$iL_C = iL_\xi + iL_{C_v} + iL_{G1} + iL_{v\xi_1} + iL_{G2},$$

where the terms are defined as

$$\begin{aligned} iL_\xi &\equiv \sum_{k=1}^2 v_{\xi_k} \frac{\partial}{\partial \xi_k} \\ iL_{C_v} &\equiv - \sum_{i=1}^N v_{\xi_1} \mathbf{v}_i \cdot \nabla_{\mathbf{v}_i} \\ iL_{G1} &\equiv G_1 \frac{\partial}{\partial v_{\xi_1}} \\ iL_{v\xi_1} &\equiv - (v_{\xi_1} v_{\xi_2}) \frac{\partial}{\partial v_{\xi_1}} \\ iL_{G2} &\equiv G_2 \frac{\partial}{\partial v_{\xi_2}}. \end{aligned}$$

The factorization for the Trotter equation that we use is⁵

$$\begin{aligned} e^{(iL_C \Delta t/2)} &= e^{(iL_{G2} \Delta t/4)} e^{(iL_{v\xi_1} \Delta t/4 + iL_{G1} \Delta t/4)} \\ &\quad \times e^{(iL_\xi \Delta t/2)} e^{(iL_{C_v} \Delta t/2)} e^{(iL_{G1} \Delta t/4 + iL_{v\xi_1} \Delta t/4)} e^{(iL_{G2} \Delta t/4)} \\ &= e^{(iL_{G2} \Delta t/4)} \left[e^{(iL_{v\xi_1} \Delta t/8)} e^{(iL_{G1} \Delta t/4)} e^{(iL_{v\xi_1} \Delta t/8)} \right] \\ &\quad \times e^{(iL_\xi \Delta t/2)} e^{(iL_{C_v} \Delta t/2)} \\ &\quad \times \left[e^{(iL_{v\xi_1} \Delta t/8)} e^{(iL_{G1} \Delta t/4)} e^{(iL_{v\xi_1} \Delta t/8)} \right] e^{(iL_{G2} \Delta t/4)}. \quad (\text{L.6.18}) \end{aligned}$$

Our numerical algorithm is now fully defined by Eqs. (L.6.17) and (L.6.18). This seemingly complicated set of equations is actually relatively easy to implement in a simulation.

To see how the implementation works, we need to know how each operator works on our coordinates $\boldsymbol{\eta} = (\mathbf{r}^N, \mathbf{v}^N, \xi_1, v_{\xi_1}, \xi_2, v_{\xi_2})$. If we start at $t = 0$ with initial condition $\boldsymbol{\eta}$, the position at time $t = \Delta t$ follows from

$$e^{iL_{\text{NHC}} \Delta t} f[\mathbf{r}^N, \mathbf{v}^N, \xi_1, v_{\xi_1}, \xi_2, v_{\xi_2}].$$

Because of the Trotter expansion, we can apply each term in iL_{NHC} , sequentially. For example, if we let the first term of the Liouville operator, iL_{G2} , act on

⁵ The second factorization, indicated by $[\dots]$, is used to avoid a hyperbolic sine function, which has a possible singularity. See ref. [126] for details.

the initial state η ,

$$\begin{aligned} & \exp\left(\frac{\Delta t}{4} G_2 \frac{\partial}{\partial v_{\xi_2}}\right) f\left[\mathbf{r}^N, \mathbf{p}^N, \xi_1, v_{\xi_1}, \xi_2, v_{\xi_2}\right] \\ &= \sum_{n=0}^{\infty} \frac{(G_2 \Delta t / 4)^n}{n!} \frac{\partial^n}{\partial v_{\xi_2}^n} f\left[\mathbf{r}^N, \mathbf{p}^N, \xi_1, v_{\xi_1}, \xi_2, v_{\xi_2}\right] \\ &= f\left[\mathbf{r}^N, \mathbf{p}^N, \xi_1, v_{\xi_1}, \xi_2, v_{\xi_2} + G_2 \Delta t / 4\right]. \end{aligned}$$

This shows that the effect of iL_{G2} is to shift v_{ξ_2} without affecting the other coordinates. This gives as transformation rule for this operator:

$$e^{(iL_{G2} \Delta t / 4)} : \quad v_{\xi_2} \rightarrow v_{\xi_2} + G_2 \Delta t / 4. \quad (\text{L.6.19})$$

The operators ($iL_{v\xi_1}$ and iL_{Cv}) are of the form $\exp(ax\partial/\partial x)$; such operators give a scaling of the x coordinate:

$$\begin{aligned} \exp\left(ax \frac{\partial}{\partial x}\right) f(x) &= \exp\left(a \frac{\partial}{\partial \ln(x)}\right) f\{\exp[\ln(x)]\} \\ &= f\{\exp[\ln(x) + a]\} = f[x \exp(a)] \end{aligned}$$

If we apply this result⁶ to $iL_{v\xi_1}$, we obtain for this operator

$$\begin{aligned} & \exp\left(-\frac{\Delta t}{8} v_{\xi_2} v_{\xi_1} \frac{\partial}{\partial v_{\xi_1}}\right) f\left[\mathbf{r}^N, \mathbf{p}^N, \xi_1, v_{\xi_1}, \xi_2, v_{\xi_2}\right] \\ &= f\left[\mathbf{r}^N, \mathbf{p}^N, \xi_1, \exp\left(-\frac{\Delta t}{8} v_{\xi_2}\right) v_{\xi_1}, \xi_2, v_{\xi_2}\right], \end{aligned}$$

giving the transformation rule

$$e^{(iL_{v\xi_1} \Delta t / 8)} : \quad v_{\xi_1} \rightarrow \exp[-v_{\xi_2} \Delta t / 8] v_{\xi_1}. \quad (\text{L.6.20})$$

In a similar way, we can derive for the other terms

$$e^{(iL_{G1} \Delta t / 4)} : \quad v_{\xi_1} \rightarrow v_{\xi_1} + G_1 \Delta t / 4 \quad (\text{L.6.21})$$

$$e^{(iL_{\xi} \Delta t / 2)} : \quad \xi_1 \rightarrow \xi_1 - v_{\xi_1} \Delta t / 2 \quad (\text{L.6.22})$$

⁶ This can be generalized, giving the identity

$$\begin{aligned} \exp\left(a \frac{\partial}{\partial g(x)}\right) f(x) &= \exp\left(a \frac{\partial}{\partial g(x)}\right) f\{g^{-1}[g(x)]\} \\ &= \exp\left(a \frac{\partial}{\partial y}\right) f[g^{-1}(y)] \\ &= f\{g^{-1}[y + a]\} = f\{g^{-1}[g(x) + a]\}. \end{aligned}$$

$$\xi_2 \rightarrow \xi_2 - v_{\xi_2} \Delta t / 2 \quad (\text{L.6.23})$$

$$e^{(iL_{Cv} \Delta t / 2)} : \quad v_i \rightarrow \exp[-v_{\xi_1} \Delta t / 2] v_i. \quad (\text{L.6.24})$$

Finally, the transformation rules that are associated with iL_v and iL_r are similar to the velocity Verlet algorithm, i.e.,

$$e^{(iL_v \Delta t / 2)} : \quad \mathbf{v}_i \rightarrow \mathbf{v}_i + \mathbf{F}_i \Delta t / (2m) \quad (\text{L.6.25})$$

$$e^{(iL_r \Delta t)} : \quad \mathbf{r}_i \rightarrow \mathbf{r}_i + \mathbf{v}_i \Delta t. \quad (\text{L.6.26})$$

With these transformation rules (L.6.19)–(L.6.26) we can write down our numerical algorithm by subsequently applying the transformation rules according to the order defined by Eqs. (L.6.17) and (L.6.18). If we start with initial coordinate $\boldsymbol{\eta}(0) = (\mathbf{r}^N, \mathbf{v}^N, \xi_1, v_{\xi_1}, \xi_2, v_{\xi_2})$, we have to apply first $e^{iL_C} \boldsymbol{\eta}$. Since this operator is further factorized according to Eq. (L.6.18), the first step in our algorithm is to apply $e^{(iL_{G2} \Delta t / 4)}$. According to transformation rule (L.6.19) applying this operator on $\boldsymbol{\eta}$ gives as new state

$$v_{\xi_2}(\Delta t / 4) = v_{\xi_2} + G_2 \Delta t / 4.$$

The output of this rule is the new state on which we apply the next operator in Eq. (L.6.18), $iL_{v_{\xi_1}}$, with transformation rule (L.6.22):

$$v_{\xi_1}(\Delta t / 8) = \exp[-v_{\xi_2}(\Delta t / 4) \Delta t / 8] v_{\xi_1}.$$

The next step is to apply iL_{G1} , followed by again $iL_{v_{\xi_1}}$, etc. In this way, we continue to apply all operators on the output of the previous step.

Applying the Nosé-Hoover part of the Liouville operator changes ξ_k , v_{ξ_k} , and $\mathbf{v}_i$. The other two Liouville operators change $\mathbf{v}_i$ and $\mathbf{r}_i$. This makes it convenient to separate the algorithm into two parts in which the positions and velocities of the particles and the Nosé-Hoover chains are considered separately. An example of a possible implementation is shown in Algorithm 44. The updates of the velocities and positions, i.e., the inner terms in Eq. (L.6.17), is done with the velocity-Verlet algorithm (Eq. (7)).

L.6.2.2 The isothermal-isobaric ensemble

Similar to the canonical ensemble, we can derive a time-reversible integration scheme for simulation in the NPT ensemble. The equations of motions are given by expressions (L.6.27)–(L.6.34):

$$\dot{\mathbf{r}}_i = \frac{\mathbf{p}_i}{m_i} + \frac{p_\epsilon}{W} \mathbf{r}_i \quad (\text{L.6.27})$$

$$\dot{\mathbf{p}}_i = \mathbf{F}_i - \left(1 + \frac{d}{dN}\right) \frac{p_\epsilon}{W} \mathbf{p}_i - \frac{p_{\xi_1}}{Q_1} \mathbf{p}_i \quad (\text{L.6.28})$$

$$\dot{V} = dV p_\epsilon / W \quad (\text{L.6.29})$$

Algorithm 44 (Equations of motion: Nosé-Hoover)

```

function integrate      integrate equations of motion
                        for Nosé-Hoover thermostat

chain(K)
vel_verlet(K)
chain(K)
end function

```

Specific Comments (for general comments, see p. 7)

1. This function solves the equations of motion for a single time step Δt using the Trotter expansion (Eqs. (L.6.17) and (L.6.18)).
2. In function **chain** we apply $e^{iL_C \Delta t/4}$ to the current state (see Algorithm 45).
3. In function **pos_vel** we apply $e^{(iL_r + iL_p)\Delta t}$ to the current state (see Algorithm 7).
4. K is the total kinetic energy: it is modified to impose a canonical distribution on K .

$$\dot{p}_\epsilon = dV(P_{\text{int}} - P_{\text{ext}}) + \frac{1}{N} \sum_{i=1}^N \frac{\mathbf{p}_i^2}{m_i} - \frac{p_{\xi_1}}{Q_1} p_\epsilon \quad (\text{L.6.30})$$

$$\dot{\xi}_k = \frac{p_{\xi_k}}{Q_k} \quad \text{for } k = 1, \dots, M \quad (\text{L.6.31})$$

$$\dot{p}_{\xi_1} = \sum_{i=1}^N \frac{\mathbf{p}_i^2}{m_i} + \frac{p_\epsilon^2}{W} - (dN + 1)k_B T - \frac{p_{\xi_2}}{Q_2} p_{\xi_1} \quad (\text{L.6.32})$$

$$\dot{p}_{\xi_k} = \frac{p_{\xi_{k-1}}^2}{Q_{k-1}} - k_B T - \frac{p_{\xi_{k+1}}}{Q_{k+1}} p_{\xi_k} \quad \text{for } k = 2, \dots, M-1 \quad (\text{L.6.33})$$

$$\dot{p}_{\xi_M} = \frac{p_{\xi_{M-1}}^2}{Q_{M-1}} - k_B T. \quad (\text{L.6.34})$$

To derive a time-reversible numerical integration scheme to solve the equations of motion, we use again the Liouville approach.

A state is characterized by the variables $\boldsymbol{\eta} = (\mathbf{r}^N, \mathbf{p}^N, \epsilon, p_\epsilon, \xi^M, p_\xi^M)$. The Liouville operator is defined by

$$iL_{NPT} \equiv \dot{\boldsymbol{\eta}} \frac{\partial}{\partial \boldsymbol{\eta}}.$$

For a chain of length $M = 2$, using $\mathbf{p}_i = m_i \mathbf{v}_i (\neq m_i \dot{\mathbf{r}}_i)$, $p_{\xi_k} = Q_k v_{\xi_k}$, $\epsilon = (\ln V)/d$, and $p_\eta = W v_\eta$, the Liouville operator for these equations of motion can be written as

$$iL_{NPT} = iL_r + iL_v + iL_{CP},$$

Algorithm 45 (Equations of motion of NH chain)**function** chain(K)apply Eq. (L.6.18)
to the current position

G2=(Q1*vxil*vxil-temp)

vxi2=vxi2+G2*delt4

Update v_{ξ_2} using Eq. (L.6.19)

vxil=vxil*exp(-vxi2*delt8)

Update v_{ξ_1} using Eq. (L.6.20)

G1=(2*K-L*temp)/Q1

vxil=vxil+G1*delt4

Update v_{ξ_1} using Eq. (L.6.21)

vxi1=vxi1*exp(-vxi2*delt8)

Update v_{ξ_1} using Eq. (L.6.20)

xil=xil+vxil*delt2

Update ξ_1 using Eq. (L.6.22)

xi2=xi2+vxi2*delt2

Update ξ_2 using Eq. (L.6.23)

s=exp(-vxil*delt2)

Scale factor in Eq. (L.6.24)

for 1 ≤ i ≤ npart **do**

v(i)=s*v(i)

rescale $\mathbf{v}_i$ using Eq. (L.6.24)**enddo**

K=K*s*s

rescale kinetic energy

vxil=vxil*exp(-vxi2*delt8)

Update v_{ξ_1} using Eq. (L.6.20)

G1=(2*K-L*temp)/Q1

vxil=vxil+G1*delt4

Update v_{ξ_1} using Eq. (L.6.21)

vxi1=vxi1*exp(-vxi2*delt8)

Update v_{ξ_1} using Eq. (L.6.20)

G2=(Q1*vxil*vxil-temp)/Q2

vxi2=vxi2+G2*delt4

Update v_{ξ_2} using Eq. (L.6.19)**end function****Specific Comments** (for general comments, see p. 7)

1. In this function temp is the imposed temperature, delt = Δt , delt2 = $\Delta t/2$, delt4 = $\Delta t/4$, and delt8 = $\Delta t/8$.
2. K is the total kinetic energy.

in which we define the operators

$$iL_r = \sum_{i=1}^N (\mathbf{v}_i + v_\epsilon \mathbf{r}_i) \cdot \nabla_{\mathbf{r}_i} + v_\epsilon \frac{\partial}{\partial \epsilon}$$

$$iL_v = \sum_{i=1}^N \left[\frac{\mathbf{F}_i(\mathbf{r})}{m_i} \right] \cdot \nabla_{\mathbf{v}_i}$$

$$iL_{CP} = - \sum_{i=1}^N v_{\xi_1} \mathbf{v}_i \cdot \nabla_{\mathbf{v}_i} + \sum_{k=1}^M v_{\xi_k} \frac{\partial}{\partial \xi_k} + \sum_{k=1}^{M-1} (G_k - v_{\xi_k} v_{\xi_{k+1}}) \frac{\partial}{\partial v_{\xi_k}}$$

$$+ G_M \frac{\partial}{\partial v_{\xi_M}} - \left(1 + \frac{1}{N}\right) \sum_{i=1}^N v_\epsilon \mathbf{v}_i \cdot \nabla_{\mathbf{v}_i} + (G_\epsilon - v_\epsilon v_{\xi_1}) \frac{\partial}{\partial v_\epsilon}$$

with

$$\begin{aligned} G_1 &= \frac{1}{Q} \left[\sum_{i=1}^N m_i \mathbf{v}_i^2 + W v_\epsilon^2 - (N_f + 1) k_B T \right] \\ G_k &= \frac{1}{Q_k} \left(Q_{k-1} v_{\xi_{k-1}}^2 - k_B T \right) \\ G_\epsilon &= \frac{1}{W} \left[\left(1 + \frac{1}{N}\right) \sum_{i=1}^N m_i \mathbf{v}_i^2 + \sum_{i=1}^N \mathbf{r}_i \cdot \mathbf{F}_i(\mathbf{r}) m_i - dV \frac{\partial U(\mathbf{r}, V)}{\partial V} - dP_{\text{ext}} V \right]. \end{aligned}$$

An appropriate Trotter equation for the equations of motion is [126]

$$\mathbf{e}^{(iL_{NPT} \Delta t)} = \mathbf{e}^{(iL_{CP} \Delta t/2)} \mathbf{e}^{(iL_v \Delta t/2)} \mathbf{e}^{(iL_r \Delta t)} \mathbf{e}^{(iL_v \Delta t/2)} \mathbf{e}^{(iL_{CP} \Delta t/2)} + \mathcal{O}(\Delta t^3). \quad (\text{L.6.35})$$

The operator iL_{CP} has to be further factorized:

$$iL_{CP} = iL_\xi + iL_{C_v} + iL_{G_\epsilon} + iL_{v_\epsilon} + iL_{G_1} + iL_{v_{\xi_1}} + iL_{G_2},$$

where the terms are defined as

$$\begin{aligned} iL_\xi &\equiv \sum_{k=1}^2 v_{\xi_k} \frac{\partial}{\partial \xi_k} \\ iL_{C_v} &\equiv - \sum_{i=1}^N \left[v_{\xi_1} + \left(1 + \frac{d}{dN}\right) \right] \mathbf{v}_i \cdot \nabla_{\mathbf{v}_i} \\ iL_{G_\epsilon} &\equiv G_\epsilon \frac{\partial}{\partial v_\epsilon} \\ iL_{v_\epsilon} &\equiv - (v_{\xi_1} v_\epsilon) \frac{\partial}{\partial v_\epsilon} \\ iL_{G_1} &\equiv G_1 \frac{\partial}{\partial v_{\xi_1}} \\ iL_{v_{\xi_1}} &\equiv - (v_{\xi_1} v_{\xi_2}) \frac{\partial}{\partial v_{\xi_1}} \\ iL_{G_2} &\equiv G_2 \frac{\partial}{\partial v_{\xi_2}}. \end{aligned}$$

The Trotter expansion of the term iL_C is

$$\mathbf{e}^{(iL_{CP} \Delta t/2)} = \mathbf{e}^{(iL_{G_2} \Delta t/4 + iL_{v_{\xi_1}} \Delta t/4)} \mathbf{e}^{(iL_{G_1} \Delta t/4)} \mathbf{e}^{(iL_{G_\epsilon} \Delta t/4 + iL_{v_\epsilon} \Delta t/4)}$$

$$\begin{aligned}
& \times e^{(iL_\xi \Delta t/2)} e^{(iL_{Cv} \Delta t/2)} \\
& \times e^{(iL_{v\epsilon} \Delta t/4 + iL_{G\epsilon} \Delta t/4)} e^{(iL_{G1} \Delta t/4 + iL_{v\xi_1} \Delta t/4)} e^{(iL_{G2} \Delta t/4)} \\
& = e^{(iL_{G2} \Delta t/4)} \left[e^{(iL_{v\xi_1} \Delta t/8)} e^{(iL_{G1} \Delta t/4)} e^{(iL_{v\xi_1} \Delta t/8)} \right] \\
& \times \left[e^{(iL_{v\epsilon} \Delta t/8)} e^{(iL_{G\epsilon} \Delta t/4)} e^{(iL_{v\epsilon} \Delta t/8)} \right] \\
& \times e^{(iL_\xi \Delta t/2)} e^{(iL_{Cv} \Delta t/2)} \left[e^{(iL_{v\epsilon} \Delta t/8)} e^{(iL_{G\epsilon} \Delta t/4)} e^{(iL_{v\epsilon} \Delta t/8)} \right] \\
& \times \left[e^{(iL_{v\xi_1} \Delta t/8)} e^{(iL_{G1} \Delta t/4)} e^{(iL_{v\xi_1} \Delta t/8)} \right] e^{(iL_{G2} \Delta t/4)}. \quad (\text{L.6.36})
\end{aligned}$$

Similar to the *NVT* version, the transformation rules of the various operators can be derived and translated into an algorithm. Such an algorithm is presented in ref. [126].

L.7 Ewald summation in a slab geometry

In Chapter 11, we discussed the treatment of long-range interactions in three-dimensional systems. For some applications, one is interested in a system that is finite in one dimension and infinite in the other two dimensions. Examples of such systems are fluids adsorbed in slit-like pores or monolayers of surfactants.

Special techniques are required to compute long-range interactions in such inhomogeneous systems. The most straightforward solution would be to use the same approach as for the three-dimensional Ewald summation, but restrict the reciprocal-space sum to vectors in the x, y directions [740,741]. The energy we wish to calculate is

$$U_{\text{Coul}} = \frac{1}{2} \sum_{i,j=1}^N \sum_{\mathbf{n}}' \frac{q_i q_j}{|\mathbf{r}_{ij} + \mathbf{n}|},$$

where the summation over $\mathbf{n} = (L_x n_x, L_y n_y, 0)$ indicates that periodicity is only imposed in the x, y directions. As in the ordinary Ewald summation, the prime indicates that for cell $(0, 0, 0)$ the terms $i = j$ should be omitted. We have a two-dimensional periodicity in the x, y directions for which we can use the Fourier representation. The resulting expression for the energy is [742]

$$\begin{aligned}
U_{\text{Coul}} = \frac{1}{2} \sum_{i,j=1}^N q_i q_j \left[\sum_{\mathbf{n}}' \frac{\text{erfc}(\alpha |\mathbf{r}_{ij} + \mathbf{n}|)}{|\mathbf{r}_{ij} + \mathbf{n}|} + \frac{\pi}{L^2} \sum_{h>0} \cos(\mathbf{h} \cdot \mathbf{r}_{ij}) F(h, z_{ij}, \alpha) \right. \\
\left. - g(z_{ij}, \alpha) \right] - \frac{\alpha}{\sqrt{\pi}} \sum_{i=1}^M q_i^2, \quad (\text{L.7.1})
\end{aligned}$$

where $\mathbf{h} \equiv (2\pi m_x/L_x, 2\pi m_y/L_y, 0)$ denotes a reciprocal lattice vector, z_{ij} is the distance between two particles in the z direction, and α is the screening

parameter. The function $F(h, z_{ij}, \alpha)$

$$F(h, z, \alpha) = \frac{\exp(hz)\operatorname{erfc}[\alpha z + h/(2\alpha)] + \exp(-hz)\operatorname{erfc}[-\alpha z + h/(2\alpha)]}{2h} \quad (\text{L.7.2})$$

corrects for the inhomogeneity in the non-periodic direction. If the system is truly two-dimensional, this term takes a simpler form. The function $g(z, \alpha)$

$$g(z, \alpha) = z \operatorname{erf}(\alpha z) + \exp\left[-(z\alpha)^2\right]/(\alpha\sqrt{\pi}) \quad (\text{L.7.3})$$

is an additional self-term of charge interactions in the central cell that must be subtracted from the reciprocal-space sum. In a neutral system with all particles in the plane $z = 0$ this term disappears. The last term in Eq. (L.7.1) is the same self-term that appears in the normal Ewald summation (11.2.21). The details of the derivation can be found in refs. [740–744].

From a computational point of view, Eq. (L.7.1) is inconvenient. Unlike the three-dimensional case, the double sum over the particles in the Fourier part of Eq. (L.7.1) can, in general, not be expressed in terms of the square of a single sum. This makes the calculation much more expensive than its three-dimensional counterpart. Several methods have been developed to increase the efficiency of the evaluation of the Ewald sum for slab geometries. Spohr [745] showed that the calculation can be made more efficient by the use of a look-up table combined with an interpolation scheme and the long-distance limit given by Eq. (L.7.6).

Hautman and Klein [746] considered the case in which the deviation of the charge distribution from a purely two-dimensional system is small. For such a system one can introduce a Taylor expansion in z , to separate the in-plane contributions x, y in $1/\sqrt{x^2 + y^2 + z^2}$ from the out-of-plane contributions. Using this approach, Hautman and Klein derived an expression in which the Fourier contribution can again be expressed in terms of sums over single particles. However, unless the ratio $z/\sqrt{x^2 + y^2} \ll 1$, the Taylor expansion converges very poorly. Therefore the applicability of this method is limited to systems in which all charges are close to a single plane. An example of such a system would be a self-assembled monolayer in which only the head groups carry a charge [746].

An obvious idea would be to use the three-dimensional Ewald summation by placing a slab of vacuum in between the periodic images (see Fig. L.8). Spohr has shown [745], however, that even with a slab that is four times the distance between the charges one does not obtain the correct limiting behavior (see Example 25). The reason is that a periodically repeated slab behaves like a stack of parallel plate capacitors. If the slab has a net dipole moment, then there will be spurious electric fields between the periodic images of the slab. More importantly, the usual assumption that the system is embedded in a conducting sphere does not correctly account for the depolarizing field that prevails in a system with a (periodic) slab geometry. Yeh and Berkowitz [747] have shown

that one can add a correction term to obtain the correct limiting behavior in the limit of an infinitely thin slab. In the limit of an infinitely thin slab in the z direction, the force on a charge q_i due to the depolarizing field is given by [748]

$$F_z = -\frac{4\pi q_i}{V} M_z, \quad (\text{L.7.4})$$

and the total electrostatic energy due to this field is

$$U_c = -\frac{2\pi}{V} M_z^2, \quad (\text{L.7.5})$$

where M_z is the net dipole moment of the simulation cell in the z direction

$$M_z = \sum_{i=1}^N q_i z_i.$$

If the slab is not infinitely thin compared to the box dimensions, higher-order correction terms have to be added. However, Yeh and Berkowitz [747] have shown that the lowest-order correction is sufficient if the spacing between the periodically repeated slabs is three to five times larger than the thickness of the slab (see also Crozier et al. [749]).

Illustration 25 (Ewald in slab). To illustrate the difficulties that arise when using the Ewald approach for computing long-range forces in a slab geometry, Spohr and co-workers [745,749] considered a simple example of two point charges: a charge $+q$ at $(0, 0, z)$ and a charge $-q$ at $(0, 0, 0)$. The system is finite in the z direction and periodic in the x, y directions with box sizes L_x and L_y (see Fig. L.7). Because of the periodic boundary conditions, the system forms two “sheets” of opposite charge.

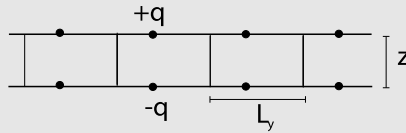


FIGURE L.7 A system containing two point charges at positions $(0, 0, 0)$ and $(0, 0, z)$; because of the periodic boundary conditions in the x and y directions, two oppositely charge “sheets” are formed. There are no periodic boundary conditions in the z direction.

In the limit $z \rightarrow \infty$, the distance between the periodic images of the charge is small compared to the distance between the sheets. We can therefore assume a uniform charge density $q/(L_x L_y)$ on each sheet. In this limit, the force acting between the two particles is given by

$$F_z = \frac{2\pi q^2}{L_x L_y}. \quad (\text{L.7.6})$$

It is instructive to compare the various methods to compute the long-range interactions in this geometry. The true forces are given by the two-dimensional Ewald summation, and we can compare the following methods:

- *Two-Dimensional Ewald summation* is the “exact” solution to this problem.
- *Bare Coulomb Potential*, we simply assume that the periodic images do not exist (or give a zero contribution). The resulting forces follow Coulomb law.
- *Truncated and Shifted Coulomb Potential*, in this method, it is assumed that beyond $r_c = 9$ the potential is zero. To remove the discontinuity at $r = r_c$ the potential is shifted as well (see section 3.3.2.2).
- *Three-Dimensional Ewald Summation*, in this approximation a layer of vacuum is added. The total system (vacuum plus slab) is seen as a normal periodic three-dimensional system (see Fig. L.8) for which the three-dimensional Ewald summation (see Eq. (11.2.24)) is used. To study the effect of the thickness of the slab of vacuum, two systems are considered, one with $L_z = 3L_x$ and a larger one with $L_z = 5L_x$.

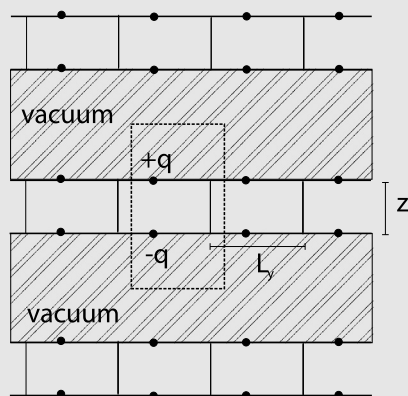


FIGURE L.8 The system of Fig. L.7 artificially made periodic in the z direction by adding a slab of vacuum.

- *3-Dimensional Ewald Summation with Correction Term*, this method is similar to the previous one; i.e., the normal three-dimensional Ewald summation is used with an additional slab of vacuum, except that now we correct for the spurious dipolar interactions.

In Fig. L.9 we compare the various approximations with the true two-dimensional solution. The bare Coulomb potential and the shifted and truncated Coulomb potential both give a zero force in the limit $z \rightarrow \infty$ and therefore do not lead to the correct limiting behavior. Although the three-dimensional Ewald summation gives a better approximation of the correct

solution, it still has the incorrect limiting behavior for both a small and a large added slab of vacuum. The corrected three-dimensional Ewald summation, however, does reproduce the correct solution, for both a slab of vacuum of $3L_x$ and that of $5L_x$.

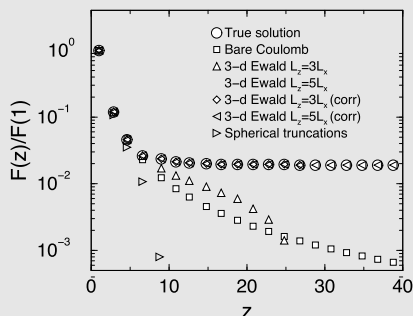


FIGURE L.9 Comparison of various methods for approximating the long-range interaction for two charges of the slab geometry shown in Fig. L.7.

L.8 Special configurational-bias Monte Carlo cases

L.8.1 Generation of branched molecules

The generation of trial configurations for branched alkanes requires some care. Naively, one might think that it is easiest to grow a branched alkane atom by atom. However, at the branchpoint we have to be careful. Suppose we have grown the backbone shown in Fig. L.10 and we now have to add the branches b_A and b_B . The total bond-bending potential has three contributions, given by

$$u_{\text{bend}} = u_{\text{bend}}(c_1, c_2, b_A) + u_{\text{bend}}(c_1, c_2, b_B) + u_{\text{bend}}(b_A, c_2, b_B).$$

Vlugt [750] pointed out that, because of the term $u_{\text{bend}}(b_A, c_2, b_B)$, it is better not to generate the positions of b_A and b_B independently. Suppose that we would try to do this anyway. We would then generate the first trial position, b_A , according to

$$P(b_A) \propto \exp[-\beta u_{\text{bend}}(c_1, c_2, b_A)],$$

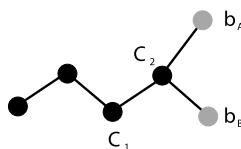


FIGURE L.10 Growth of a branched alkane.

next, we would generate the second trial position, b_B , using

$$P(b_B|b_A) \propto \exp\{-\beta[u_{\text{bend}}(c_1, c_2, b_B) + u_{\text{bend}}(b_A, c_2, b_B)]\},$$

where $P(b_B|b_A)$ denotes the probability of generating b_B for a given position of segment b_A . However, if we would generate both positions at the same time, then the probability is given by

$$P(b_A, b_B) \propto \exp\{-\beta[u_{\text{bend}}(c_1, c_2, b_A) + u_{\text{bend}}(c_1, c_2, b_B) + u_{\text{bend}}(b_A, c_2, b_B)]\}.$$

The two schemes are only equivalent if

$$P(b_A, b_B) = P(b_B|b_A)P(b_A).$$

In general, this equality does not hold. To see this, compare the probability of generating configuration b_A for the two schemes. This probability is obtained by integrating over all orientations b_B . If both chains are inserted at the same time, we find that

$$\begin{aligned} P(b_A) &= \int db_B P(b_A, b_B) \\ &\propto \exp[-\beta u_{\text{bend}}(c_1, c_2, b_A)] \\ &\quad \times \int db_B \exp\{-\beta[u_{\text{bend}}(c_1, c_2, b_A) + u_{\text{bend}}(b_A, c_2, b_B)]\}. \end{aligned}$$

For the sequential scheme, we would have obtained

$$\begin{aligned} P(b_A) &= \int db_B P(b_B|b_A)P(b_A) \\ &= P(b_A) \\ &\propto \exp[-\beta u_{\text{bend}}(c_1, c_2, b_A)] \end{aligned}$$

as, in this scheme, segment b_A is inserted before segment b_B . Therefore the probability $P(b_A)$ cannot depend on b_B .

We can now easily see that if we use a model in which the two branches are equivalent, for example, isobutane, the sequential scheme does not generate equivalent *a priori* distributions for the two branches. Of course, the generation of trial segments is but one step in the **CBMC** scheme. Any bias introduced at this stage can be removed by incorporating the ratio of the true and the biased distributions in the acceptance criterion. However, the resulting algorithm may be inefficient. Vlught et al. [538] have shown that simply ignoring the bias introduced by the “sequential” scheme will result in small, but noticeable, errors in the distribution of the bond angles.

The insertion of two segments at the same time is less efficient than sequential insertion, several strategies have been proposed to increase the efficiency of the simultaneous generation of branches.

For molecules in which the bond-bending potential has three contributions (as in the example above), the simplest scheme is to generate two random vectors on a sphere and use the conventional rejection scheme to generate configurations with a probability proportional to their Boltzmann weight [751]. One can also use this approach for more complex potentials that include torsion. If the random generation of trial directions becomes inefficient, it may be replaced by a simple Monte Carlo scheme [538].

For some intramolecular potential, it may even be necessary to add more than two atoms at the same time to ensure a proper *a priori* distribution of added segments. In fact, for some molecules that have multiple torsional angles, such as 2,3-dimethylbutane, this approach would imply that all atoms have to be added at the same time. To avoid such many-particle insertions, Martin and Siepmann [752] developed a scheme similar to the multiple-first-bead algorithm (see section 12.5).

The idea is to use a random insertion to generate several trial positions and to use a CBMC scheme to select acceptable candidates using the internal energies only. These configurations that are distributed according to the correct intramolecular Boltzmann weight will subsequently be used in another CBMC scheme that involves more expensive external energy calculations.

To see how this approach works, assume that we have a model with internal interactions given by u^{int} . A single segment is added using the following steps:

1. First generate a set of n_{int} random trial positions and for each position compute the internal energy, $U^{\text{int}}(i)$, and calculate the Rosenbluth factor, W associated with this internal energy

$$W^{\text{int}}(n) = \sum_{j=1}^{n_{\text{int}}} \exp \left[-\beta U^{\text{int}}(j) \right].$$

A possible orientation is then selected using

$$P^{\text{int}}(j) = \frac{\exp \left[-\beta U^{\text{int}}(j) \right]}{W^{\text{int}}(n)}.$$

2. Step 1 is repeated to generate k trial positions, which are then fed into the conventional CBMC scheme to compute the Rosenbluth factor, W , using the external potential $W^{\text{ext}}(n)$.
3. A similar method is used for the old configuration, giving $W^{\text{int}}(o)$ and $W^{\text{ext}}(o)$.
4. A move is accepted using

$$\text{acc}(o \rightarrow n) = \min \left(1, \frac{W^{\text{int}}(n) W^{\text{ext}}(n)}{W^{\text{int}}(o) W^{\text{ext}}(o)} \right).$$

Depending on the details of the potential, further refinements are possible. One can, for instance, separate the bond-bending potential and the torsion potential. This would imply three nested CBMC steps giving three different Rosenbluth factors. For more details see [752].

L.8.2 Rebridging Monte Carlo

If we model a realistic polymer or peptide we have to include bond-bending and torsional potentials. Suppose that we rotate in the interior of a polymer a randomly selected torsional angle by an amount $\Delta\phi$. If we would keep all other torsional angles of the remainder of the chain fixed, a tiny change of this torsional angle would lead to a large displacement of the last atom of the chain. If, on the other hand, one would only displace the neighboring atoms, the intramolecular interactions of the chain would increase significantly, again limiting the maximum rotation. We would like to ensure that the rotation affects only a small part of the interior of the chain and that it results in a redistribution of atoms that does not result in a large increase in the intramolecular energy. *Concerted rotation* and *rebridging* and *end-bridging* Monte Carlo are schemes that have been developed by Theodorou and co-workers [99,517,753] to perform such Monte Carlo moves.

In Fig. L.11 the rebridging problem is sketched schematically. Suppose we give the atoms 1 and 5 a new position by a random rotation of the driver angles ϕ_0 and ϕ_7 . Assume that all bond lengths and bond angles have a prescribed value, for example, their equilibrium value or any other specified value. The rebridging problem is to find all possible conformations of the trimer consisting of the atoms 2, 3, and 4 that rebridge the new positions of atoms 1 and 5 given the constraints of the prescribed bond lengths and angles. Wu and Deem [754] have shown that for the rebridging problem an analytical solution exists and that the maximum number of solutions is strictly limited to 16. Alternatively, in refs. [517,753] it is shown how to numerically locate *all* solutions of the rebridging problem.

Suppose that we have all solutions to the rebridging problem, either by the analytical solution of Wu and Deem or by the numerical scheme of Theodorou and co-workers. The next step is to use this in a Monte Carlo scheme. The scheme that we discuss here is only valid for the interior segments of a polymer. For the ends of the chains, a slightly different scheme has to be used [99]:

1. The present conformation of the polymer is denoted by o . We generate the new configuration of the polymer, n , by selecting an atom and a direction (forward or backward) at random. This defines the atom's pair 1 and 5. These atoms are given new positions $1'$ and $5'$ by performing a random rotation around bonds $-1, 0$ and $6, 7$, respectively (see Fig. L.11).
2. Solve the rebridging problem to locate all possible conformations of the trimer that bridge the new positions of atoms $1'$ and $5'$. The total number of conformations is denoted by N_n and out of these we select one confor-

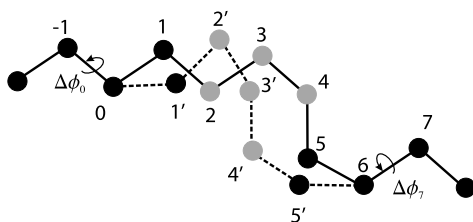


FIGURE L.11 Schematic drawing of the rebridging Monte Carlo scheme. Suppose we give the atoms 1 and 5 a new position, for example, by rotating around the $-1, 0$ and $6, 7$ bonds by angles $\Delta\phi_0$ and $\Delta\phi_7$, respectively. If we would not change the positions of the trimer, consisting of the gray atoms 2, 3, and 4, the intramolecular energy would increase significantly. The rebridging problem is to find a new conformation of the trimer with the same bond length and bond angle as the old conformation that bridges the new positions $1'$ and $5'$.

mation, say n , at random.⁷ If no such conformation is found the move is rejected.

3. For the old conformation, we also locate all possible conformation, i.e., we solve the rebridging problem to locate the conformations of the trimer that bridge the old positions of atoms 1 and 5. This number of conformation is denoted by N_o .
4. In the rebridging scheme, we use a dihedral angle ϕ to generate a new configuration. This implies a temporary change of coordinate system; a Jacobian is associated with this change. In general, this Jacobian is not equal to 1. This Jacobian has to be taken into account in the acceptance rules [99]. The equations for this Jacobian can be found in refs. [99,753,754]. Here, we assume that these determinants for the old and new conformation have been calculated and are denoted by $J(o)$ and $J(n)$, respectively.
5. Of the new and old conformations the energies are calculated, $U(o)$ and $U(n)$, respectively.
6. The new conformation is accepted with a probability

$$\text{acc}(o \rightarrow n) = \min \left(1, \frac{\exp[-\beta U(n)] J(n)/N_n}{\exp[-\beta U(o)] J(o)/N_o} \right).$$

In refs. [99,753] the proof is given that this rebridging scheme obeys detailed balance and samples a Boltzmann distribution.

The reason it is important to find *all* solutions to the rebridging problem is to ensure detailed balance. Suppose that the determinants of the Jacobians are one and the energies are zero, then without the terms $1/N_n$ and $1/N_o$ the acceptance probability would be one for all possible new conformations. Suppose that we have a single solution for the new conformation, $N_n = 1$, and for the old conformation $N_o = 2$. Without the correction, we would violate detailed balance since

⁷ One could use the configurational-bias Monte Carlo scheme as an alternative for the random selection.

the *a priori* probability of the reverse move is only a half. Pant and Theodorou [753] have developed an alternative scheme in which one has to find only a single rebridging conformation, which is the first solution of their numerical scheme. To ensure detailed balance one has to check that the old conformation should also be the first solution to which the numerical scheme converges.

One can also use the rebridging scheme to connect atoms of *different* chains. The idea of end-bridging Monte Carlo is to alter the connectivity of the chain by bridging atoms from different chains. The simplest form is to rebridge a chain end to an interior segment of another chain with a trimer. Such an end-bridging Monte Carlo move induces a very large jump in configuration space. An important aspect of such an end-bridging move is, however, that it alters the chain lengths of the two chains. Therefore, such a move cannot be used if it is important to keep the chain length fixed. However, in most practical applications of polymers, one does not have a single chain length but a distribution of chain lengths. Pant and Theodorou [753] have shown that the resulting chain length distribution resembles a truncated Gaussian distribution.

One can envision a chain length distribution as a mixture of a very large number of components, each component characterized by its chain length l . Imposing the chain length distribution is equivalent to imposing the chemical potentials of the various components. This suggests that we could combine these end-bridging moves with the *semigrand ensemble* simulation technique (see section 6.5.4) to determine whether a change of the polymer length should be accepted.

In principle, one can use two rebridging moves for the interior segments of two chains. This would allow us to perform moves in which the total chain length remains constant. Whether in practice such a scheme will work depends on the probability that two segments of different chains with the *same* number of end segments connected to it are sufficiently close to each other.

Tests show that the rebridging method is very efficient for polymer melts with chain lengths up to C₃₀. For chains up to C₇₀ rebridging Monte Carlo still samples the local structure efficiently, but fails to sample chain characteristics at larger length scales such as the end-to-end vector. End-bridging Monte Carlo effectively relaxes chains up to C₅₀₀ [517]. Another important application of rebridging Monte Carlo is the possibility of simulating cyclic molecules. This application is illustrated by Wu and Deem in their study of cis/trans isomerization of proline-containing cyclic peptides [754,755].

L.8.3 Gibbs-ensemble simulations

In section 6.6, the Gibbs-ensemble technique was introduced as an efficient tool for simulating vapor-liquid phase equilibria. One of the Monte Carlo steps in the Gibbs-ensemble technique is the transfer of molecules between the liquid phase and gas phase. For long-chain molecules, this step, if carried out completely randomly, results in a prohibitively low acceptance of particle exchanges. Therefore, the Gibbs-ensemble technique used to be limited to systems containing

atoms or small molecules. However, by combining the Gibbs-ensemble method with configurational-bias Monte Carlo, the method can be made to work for much longer chain molecules.

Algorithm

Let us consider a continuum system with strong intramolecular interactions. In section 12.2.3 it is shown that for such a system, it is convenient to separate the potential energy into two contributions: the bonded intramolecular energy (U^{bond}) and the “external” energy (U^{ext}) that contains the intermolecular interactions and the nonbonded intramolecular interactions. As in the original implementation of the Gibbs ensemble, we attempt to exchange a molecule between the two boxes. However, while in section 6.6.1 the molecules were inserted at random, we now use the following procedure to grow a molecule atom by atom in a randomly selected box. Let us assume this is box 1 with volume V_1 . The number of particles in this box is denoted by n_1 .

1. The first atom is inserted at a random position, and the (external) energy $u_1^{\text{ext}}(n)$ is calculated together with

$$w_1^{\text{ext}}(n) = k \exp[-\beta u_1^{\text{ext}}(n)]. \quad (\text{L.8.1})$$

2. To insert the next atom i , k trial orientations are generated. The set of k trial orientations is denoted by $\{\mathbf{b}\}_k = \mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k$. These orientations are not generated at random but with a probability that is a function of the bonded part of the intramolecular energy:

$$p_i^{\text{bond}}(\mathbf{b}_n) = C \exp[-\beta u_i^{\text{bond}}(\mathbf{b}_n)]. \quad (\text{L.8.2})$$

Of each of these trial orientations the external energy is calculated $u_i^{\text{ext}}(\mathbf{b}_j)$ together with the factor

$$w_i^{\text{ext}}(n) = \sum_{j=1}^k \exp[-\beta u_i^{\text{ext}}(\mathbf{b}_j)]. \quad (\text{L.8.3})$$

Out of these k trial positions, we select one with probability

$$p_i^{\text{ext}}(\mathbf{b}_n) = \frac{\exp[-\beta u_i^{\text{ext}}(\mathbf{b}_i)]}{w_i^{\text{ext}}(n)}. \quad (\text{L.8.4})$$

3. Step 2 is repeated $\ell - 1$ times until the entire molecule is grown and the Rosenbluth factor, W of the molecule can be calculated:

$$W^{\text{ext}}(n) = \prod_{i=1}^{\ell} w_i^{\text{ext}}(n). \quad (\text{L.8.5})$$

For the other box, we select a molecule at random and determine its Rosenbluth factor, using the following procedure:

1. A particle is selected at random.
2. The (external) energy of the first atom is determined $u_1^{\text{ext}}(o)$ together with

$$w_1^{\text{ext}}(o) = k \exp[-\beta u_1^{\text{ext}}(o)]. \quad (\text{L.8.6})$$

3. For the next atom, $k - 1$ trial orientations are generated with a probability given by Eq. (L.8.2). These trial orientations, together with the actual position of atom i ($\mathbf{b}_o$), form the set $\{\mathbf{b}'_j\}_k$ for which we determine the factor

$$w_i^{\text{ext}}(o) = \exp[-\beta u_i^{\text{ext}}(o)] + \sum_{j=2}^k \exp[-\beta u_i^{\text{ext}}(\mathbf{b}'_j)]. \quad (\text{L.8.7})$$

4. Step 2 is repeated $\ell - 1$ times until we have retraced the entire chain and its Rosenbluth factor can be calculated:

$$W^{\text{ext}}(o) = \prod_{l=1}^{\ell} w_l^{\text{ext}}(o). \quad (\text{L.8.8})$$

We then accept this move with probability

$$\text{acc}(o \rightarrow n) = \min \left(1, \frac{V_1(N - n_1)}{(V - V_1)(n_1 + 1)} \frac{W^{\text{ext}}(n)}{W^{\text{ext}}(o)} \right). \quad (\text{L.8.9})$$

The proof of the validity of this algorithm, again, is very similar to those shown earlier in this chapter. We, therefore, refer the interested reader to [512,756,757]. The combination of the Gibbs-ensemble technique with the configurational-bias Monte Carlo method has been used to determine the vapor-liquid coexistence curve of chains of Lennard-Jones beads [512,756] and alkanes [535,757–759]. In Illustration 26, an application of this method is described.

Illustration 26 (Critical properties of alkanes). Alkanes are thermally unstable above approximately 650 K, which makes an experimental determination of the critical point of alkanes longer than decane (C_{10}) extremely difficult. The longer alkanes, however, are present in mixtures of practical importance for the petrochemical industry. In these mixtures, the number of components can be so large that it is not practical to determine all phase diagrams experimentally. One, therefore, has to rely on predictions made by equations of state. The parameters of these equations of state are directly related to the critical properties of the pure components. Therefore, the critical properties of the long-chain alkanes are used in the design of petrochemical processes, even if they are unstable close to the critical point [760]. Unfortunately, experimen-

tal data are scarce and contradictory, and one has to rely on semi-empirical methods to estimate the critical properties [760].

Siepmann et al. [535,757] have used the combination of the Gibbs-ensemble technique and **Configurational-Bias Monte Carlo (CBMC)** to simulate vapor-liquid equilibria of the *n*-alkanes under conditions where experiments are not (yet) feasible. Phase diagrams are very sensitive to the choice of interaction potentials. Most available models for alkanes have been obtained by fitting simulation data to experimental properties of the liquid under standard conditions. In Fig. L.12, the vapor-liquid curve of octane, as predicted by some of these models, is compared with experimental data. This figure shows that the models of [761,762], which give nearly identical liquid properties, yield estimates of the critical temperature of octane that differ by 100 K. Siepmann et al. [535,757] used these vapor-liquid equilibrium data to improve the existing models.

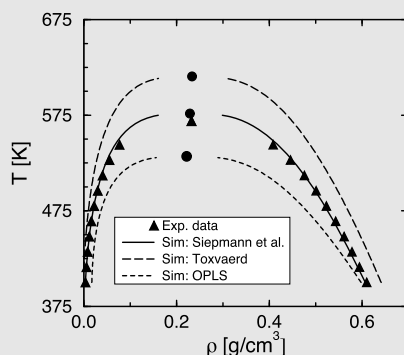


FIGURE L.12 The vapor-liquid curve of octane: comparison of Gibbs ensemble simulations using the so-called OPLS model of Jorgensen and co-workers [761], the model of Toxvaerd [762], and the model of Siepmann et al. [535,757].

In Fig. L.13, the critical temperatures and densities as predicted by the model of Siepmann et al. are plotted versus the carbon number. The simulations reproduce the experimental critical points very well. There is considerable disagreement, however, between the various experimental estimates of the critical densities. Much of our current knowledge of the critical properties of the higher alkanes is based on extrapolations of fits of the experimental data up to C_8 . The most commonly used extrapolations assume that the critical density is a monotonically increasing function of the carbon number, approaching a limiting value for the very long alkanes [760,763]. In contrast to these expectations, the experimental data of Anselme et al. [764] indicate that the critical density has a maximum for C_8 and then decreases monotonically. The data of Steele (as reported in [763]), however, do not give any evidence for such a maximum (see Fig. L.13). The simulations indicate the same trend as that observed by Anselme et al. In this context, it is interesting to note that Mooij et al. [512], Sheng et al. [765], and Escobedo and de Pablo

[766] used Monte Carlo simulations to study the vapor-liquid curve of a polymeric bead-spring model for various chain lengths. These studies also show a decrease of the critical density as a function of chain length. Such a decrease of the critical density with chain length is a general feature of long-chain molecules, as was already pointed out by Flory.

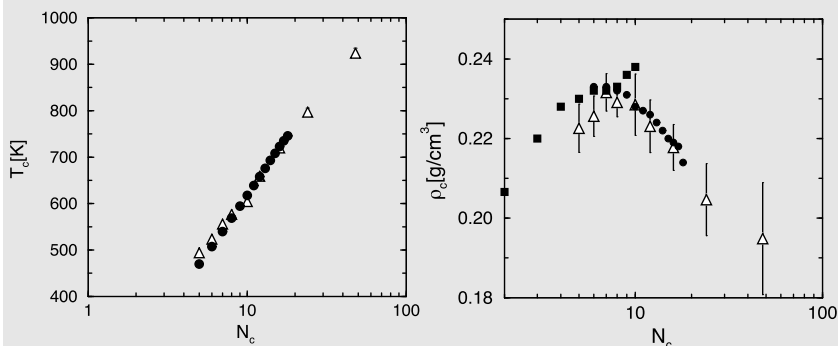


FIGURE L.13 Critical temperature T_c (left) and density ρ_c (right) as a function of carbon number N_c . The open symbols are the simulation data, and the closed symbols are experimental data.

The Gibbs ensemble technique makes it possible to compute efficiently the liquid-vapor coexistence curve of realistic models for molecular fluids. This makes it possible to optimize the parameters of the model to yield an accurate description of the entire coexistence curve rather than a single state point. It is likely, but not inevitable, that a model that describes the phase behavior correctly will also yield reasonable estimates of other properties, such as viscosity or diffusivity. Mondello and Grest have shown that this is indeed true for the diffusion coefficient of linear hydrocarbons [767,768], while Cochran, Cummings, and co-workers [769,770] found the same for the viscosity. The hydrocarbon model that was used in these studies had been optimized to reproduce experimental vapor-liquid coexistence data [535,757]. Improved force fields have since been proposed for linear alkanes [514,771,772], branched alkanes [752], alkenes [773,774], alkylbenzenes [773], and alcohols [775,776].

L.9 Recoil growth: justification of the method

The best way to arrive at the acceptance rule for the recoil growth scheme is to pretend that we actually carry out the naive brute-force calculation where we first generate the tree of *all* k^{l-1} trial conformations. We denote this tree by T_n and the *a priori* probability for generating this tree by $P_T(T_n)$. Next we test which links are “open” or “closed.” The decision whether a monomer direction is “open” or “closed” is made on the basis of the probabilities equation (12.7.1) and we denote the probability that we have a particular set O_n

of “open” monomers (and all others “closed”) by $P_O(O_n|T_n)$. Let us note the number of “open” monomers in this set by $N(O_n)$ and the number of “closed” monomers by $N(C_n)$. It is easy to see that the probability of generating this particular set is given by

$$P_O(O_n|T_n) = \prod_{j=1}^{N(O_n)} p_j^{\text{open}}(b) \prod_{k=1}^{N(C_n)} (1 - p_k^{\text{open}}(b)).$$

Finally we try to select one completely open conformation by randomly selecting, at every step, one of the “available” trial directions, i.e., a direction that is connected to (at least) one feeler that does not “die” within l_{max} steps. At every step, there are $m_i(n)$ such directions. Hence the probability of selecting a given direction is simply $1/m_i(n)$ and the total probability that a specific conformation will be selected on the previously generated tree of all possible conformations is

$$P_S(n|O_n) = \prod_{i=1}^{l-1} \frac{1}{m_i(n)}$$

if all m_i are non-zero, and

$$P_S(n|O_n) = 0$$

otherwise. The fact that the algorithm leaves out many redundant steps (*viz.* generating the “doomed” branches or checking if there is more than one open feeler in a given direction) is irrelevant for the acceptance rule. The overall probability that we generate a trial conformation n on the set O_n , $P_S(n|O_n)$, in a tree T_n is

$$P_T(T_n) \times P_O(O_n|T_n) \times P_S(n|O_n). \quad (\text{L.9.1})$$

In order to compute the acceptance probability of a trial move, we should consider the reverse move where the *old* conformation is generated. By analogy to the forward case, this probability is given by

$$P_T(T_o) \times P_O(O_o|T_o) \times P_S(o|O_o). \quad (\text{L.9.2})$$

We wish our MC scheme to obey detailed balance. However, just as in the **CBMC** case, it is easier to impose the stronger condition of *super-detailed balance*. This implies that, in the forward move, we also should consider the probability of generating a complete tree of possible conformations around the “old” conformation and the probability that a subset of all monomers on this tree is “open.” We denote the probability of generating this tree by $P'_T(T'_o)$, where the prime indicates that this is the probability of generating all branches of the old tree, except the already existing old conformation. Clearly

$$P_T(T_o) = P'_T(T'_o) \times P_{\text{gen}}(o), \quad (\text{L.9.3})$$

where $P_{\text{gen}}(o)$ denotes the probability of generating the old conformation. As in the **CBMC** scheme, we can include strong intramolecular interactions in the generation of these trial monomers (see section 12.3). $P_{\text{gen}}(o)$ will then be of the form (see section 12.3)

$$P_{\text{gen}}(o) = \left(\prod_{i=1}^l p_i^{\text{bond}}(b_o) \right). \quad (\text{L.9.4})$$

Similarly, we have to consider the probability $P'_O(O'_o|T'_o)$ that a set O'_o on this tree is “open.” Again, the prime indicates that we should not include the old conformation itself. Again, it is easy to see that

$$P_O(O_o|T_o) = \left(\prod_{i=1}^l p_i^{\text{open}}(b_o) \right) P'_O(O'_o|T'_o). \quad (\text{L.9.5})$$

The *a priori* probability of generating a trial move from o to n is then given by

$$\begin{aligned} \alpha(o \rightarrow n|T_n, O_n, T_o, O_o) \\ = P_T(T_n) \times P_O(O_n|T_n) \times P_S(n|O_n) \times P'_T(T'_o) \times P'_O(O'_o|T'_o). \end{aligned} \quad (\text{L.9.6})$$

For the reverse move $n \rightarrow o$, we can derive a similar expression:

$$\begin{aligned} \alpha(n \rightarrow o|T_n, O_n, T_o, O_o) \\ = P_T(T_o) \times P_O(O_o|T_o) \times P_S(o|O_o) \times P'_T(T'_n) \times P'_O(O'_n|T'_n). \end{aligned} \quad (\text{L.9.7})$$

In these equations, we have used the notation $(o \rightarrow n|T_n, O_n, T_o, O_o)$ to indicate that we consider a transition from o to n (or vice versa) for a *given* set of “embedding” conformations. Clearly, there are many different trees and sets of open orientations that include the same conformations n and o .

Our super-detailed balance condition now becomes

$$\begin{aligned} \mathcal{N}(o) \times \alpha(o \rightarrow n|T_n, O_n, T_o, O_o) \text{acc}(o \rightarrow n|T_n, O_n, T_o, O_o) \\ = \mathcal{N}(n) \times \alpha(n \rightarrow o|T_n, O_n, T_o, O_o) \text{acc}(n \rightarrow o|T_n, O_n, T_o, O_o). \end{aligned} \quad (\text{L.9.8})$$

All terms in this equation are known, except the acceptance probabilities. We now derive an expression for the ratio $\text{acc}(o \rightarrow n|T_n, O_n, T_o, O_o)/\text{acc}(n \rightarrow o|T_n, O_n, T_o, O_o)$. To this end, we insert Eqs. (L.9.3) and (L.9.5) (and the corresponding expressions for $P'_T(T'_n)$ and $P'_O(O'_n|T'_n)$) into our super-detailed balance condition equation (L.9.8). This leads to a huge simplification as there is a complete cancellation of all probabilities for generating “open” or “closed” monomers that do *not* belong to the new (or the old) conformation. What remains is

$$\mathcal{N}(o) \times P_{\text{gen}}(n) \left(\prod_{i=1}^l \frac{p_i^{\text{open}}(b_n)}{m_i(n)} \right) \text{acc}(o \rightarrow n|T_n, O_n, T_o, O_o)$$

$$= \mathcal{N}(n) \times P_{\text{gen}}(o) \left(\prod_{i=1}^l \frac{p_i^{\text{open}}(b_o)}{m_i(o)} \right) \text{acc}(n \rightarrow o | T_n, O_n, T_o, O_o). \quad (\text{L.9.9})$$

In order to simplify the notation, we shall assume that the trial directions are uniformly distributed, i.e., see Eq. (L.9.4), $p^{\text{bond}} = \text{constant}$. From Eq. (L.9.4) it then follows that $P_{\text{gen}}(n)$ and $P_{\text{gen}}(o)$ are identical constants.

Our expression for the ratio of the acceptance probabilities then becomes

$$\frac{\text{acc}(o \rightarrow n)}{\text{acc}(n \rightarrow o)} = \frac{\mathcal{N}(n) \prod_{i=1}^{\ell} p_i^{\text{open}}(o)/m_i(o)}{\mathcal{N}(o) \prod_{i=1}^{\ell} p_i^{\text{open}}(n)/m_i(n)}, \quad (\text{L.9.10})$$

where we have dropped the indices $T_n, O_n, \dots$. Using the definitions of $W(n)$ and $W(o)$ (Eq. (12.7.2) and below),

$$\frac{\text{acc}(o \rightarrow n)}{\text{acc}(n \rightarrow o)} = \frac{\mathcal{N}(n)W(n)}{\mathcal{N}(o)W(o)}. \quad (\text{L.9.11})$$

This is precisely the acceptance rule given by Eq. (12.7.3). This concludes our “derivation” of the recoil growth scheme. The obvious question is: how well does it perform? A comparison between CBMC and the RG algorithm was made by Consta et al. [540], who studied the behavior of Lennard-Jones chains in solution. The simulations showed that for relatively short chains ($\ell = 10$) at a density of $\rho = 0.2$, the recoil growth scheme was a factor of 1.5 faster than CBMC. For higher densities $\rho = 0.4$ and longer chains $\ell = 40$ the gain could be as large as a factor 25. This illustrates the fact that the recoil scheme is still efficient, under conditions where CBMC is likely to fail. For still higher densities or still longer chains, the relative advantage of RG would be even larger. However, the bad news is that, under those conditions, *both* schemes become very inefficient.

While the recoil growth scheme is a powerful alternative to CBMC, the RG strategy is not very useful for computing chemical potentials (see [540]). More efficient schemes for computing the chemical potential are the recursive sampling scheme and the Pruning-Enriched Rosenbluth Method (PERM) (see Chapter 10).

L.10 Overlapping distribution for polymers

Let us first consider how the basic idea behind the overlapping distribution method can be applied to the Rosenbluth insertion scheme. The simplest approach would be to consider the histogram of the potential energy change on addition or removal of a chain molecule (see section 8.6.1). However, for chain molecules, this approach differs from the original Shing-Gubbins approach in that it has little, if any, diagnostic value. For instance, if we consider the chemical potential of hard-core chain molecules, the distributions of $\Delta\mathcal{U}$ will always

overlap (namely, at $\Delta\mathcal{U} = 0$), even in the regime where the method cannot be trusted. Here, we shall describe an overlapping distribution method based on histograms of Rosenbluth weights [455]. This method will prove to be a useful diagnostic tool.

Consider again a model with internal potential energy u_{int} and external potential energy u_{ext} . In what follows, we shall compare two systems. The first, denoted by 0, contains N chain molecules ($N \geq 0$). The second system, denoted by 1, contains $N + 1$ chain molecules. In addition, both systems may contain a fixed number of other (solvent) molecules. Let us first consider system 1. Around every segment j of a particular chain molecule (say, i), we can generate $k - 1$ trial directions according to an internal probability distribution given by Eq. (10.2.19). Note that the set does *not* include the actual orientation of segment j . We denote this set of trial orientations by

$$\{\boldsymbol{\gamma}_{\text{rest}}(j)\} \equiv \prod_{j'=1}^{k-1} \{\boldsymbol{\gamma}\}_{j'},$$

where the subscript rest indicates that this set excludes the actual segment j . The probability of generating this set of trial directions is given by $P_{\text{rest}}(j)$, given by Eq. (10.2.19). Having thus constructed an umbrella of trial directions around every segment $1 \leq j \leq \ell$, we can compute the Rosenbluth weight $\mathcal{W}_i$ of molecule i . Clearly, $\mathcal{W}_i$ depends on all coordinates of the remaining N molecules (for convenience, we assume that we are dealing with a neat liquid), on the position $\mathbf{r}_i$ and conformation $\boldsymbol{\Gamma}_i$ of molecule i , and on the ℓ sets of $k - 1$ trial directions:

$$\{\boldsymbol{\Gamma}_{\text{rest}}\} \equiv \prod_{j=1}^{\ell} \{\boldsymbol{\gamma}\}_{\text{rest}}(j).$$

We now define a quantity x through

$$x \equiv \ln \mathcal{W}_i(\mathbf{Q}^{N+1}, \{\boldsymbol{\Gamma}_{\text{rest}}\}),$$

where we use $\mathbf{Q}$ to denote the translational coordinates $\mathbf{r}$ and conformational coordinates $\boldsymbol{\Gamma}$ of a molecule. Next, consider the expression for the probability density of x , $p_1(x)$:

$$p_1(x) = \frac{\int d\mathbf{Q}^{N+1} d\{\boldsymbol{\Gamma}_{\text{rest}}\} \exp[-\beta\mathcal{U}(\mathbf{Q}^{N+1})] \prod_{j=1}^{\ell} P_{\text{rest}}(j) \delta(x - \ln \mathcal{W}_i)}{Z_{N+1}},$$

where

$$Z_{N+1} = \int d\mathbf{Q}^{N+1} d\{\boldsymbol{\Gamma}_{\text{rest}}\} \exp[-\beta\mathcal{U}(\mathbf{Q}^{N+1})] \prod_{j=1}^{\ell} P_{\text{rest}}(j)$$

$$= \int \cdots \int d\mathbf{Q}^{N+1} \exp \left[-\beta \mathcal{U}(\mathbf{Q}^{N+1}) \right].$$

The second line of this equation follows from the fact that all $P_{\text{int}}(j)$ are normalized. We shall now try to relate $p_1(x)$ to an average in system 0 (i.e., the system containing only N chain molecules). To this end, we write $\mathcal{U}(\mathbf{Q}^{N+1})$ as $\mathcal{U}(\mathbf{Q}^N) = u_{\text{ex}}(\mathbf{Q}^N, \mathbf{Q}_i) + u_{\text{int}}(\mathbf{Q}_i)$. Second, we use the fact that

$$\exp[-\beta u_{\text{int}}(i)] = Z_{\text{id}} \times \prod_{j=1}^{\ell} P_{\text{int}}(j),$$

where

$$Z_{\text{id}} \equiv \int d\mathbf{r}_i \prod_{j=1}^{\ell} \exp[-\beta u_{\text{int}}(j)].$$

Our expression for $p_1(x)$ now becomes

$$\begin{aligned} p_1(x) &= \frac{Z_{\text{id}}}{Z_{N+1}} \int d\mathbf{Q}^N d\mathbf{r}_i d\{\mathbf{\Gamma}_{\text{trial}}\} \exp \left[-\beta \mathcal{U}(\mathbf{Q}^N) \right] \\ &\quad \times \prod_{j=1}^{\ell} P_{\text{trial}}(j) \exp[-\beta u_{\text{ex}}(j)] \delta(x - \ln \mathcal{W}_i). \end{aligned}$$

We use the symbol $\{\mathbf{\Gamma}_{\text{trial}}\}$ to denote the set of *all* trial segments, that is, the “umbrella” of trial directions around all segments of the chain molecule, *plus the segments themselves*. Next, every term $\exp(-\beta u_{\text{ex}}(j))$ is multiplied and divided by Z_j , defined as

$$Z_j \equiv \sum_{j'=1}^k \exp(-\beta u_{\text{ext}}(j')).$$

This allows us to write, for $p_1(x)$,

$$\begin{aligned} p_1(x) &= \frac{V Z_{\text{id}}}{Z_{N+1}} \\ &\quad \times \sum_{\text{trials}} \int d\mathbf{Q}^N d\mathbf{s}_i d\{\mathbf{\Gamma}_{\text{trial}}\} \exp \left[-\beta \mathcal{U}(\mathbf{Q}^N) \right] P_{\text{sel}}(\mathbf{Q}_i) \mathcal{W}_i \delta(x - \ln \mathcal{W}_i), \end{aligned}$$

where we have transformed from real coordinates $\mathbf{r}_i$ to scaled coordinates $\mathbf{s}_i$ by factoring out V , the volume of the system. Here, $P_{\text{sel}}(\mathbf{Q}_i)$ denotes the probability of selecting the actual conformation of the molecule from the given set of trial segments according to the rule given in Eq. (10.2.20). Finally, we multiply and

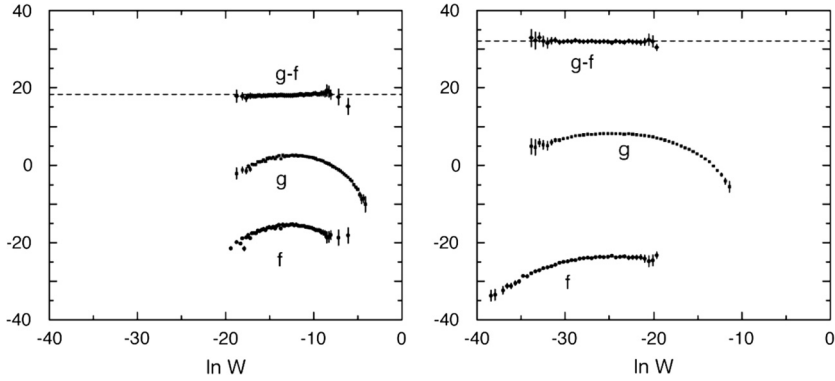


FIGURE L.14 The functions $f(\ln \mathcal{W}) \equiv p_0(\ln \mathcal{W}) + \frac{1}{2} \ln \mathcal{W}$, $g(\ln \mathcal{W}) \equiv p_1(\ln \mathcal{W}) - \frac{1}{2} \ln \mathcal{W}$ for fully flexible chains of hard sphere of length (left) $\ell = 8$ and (right) $\ell = 14$ in a hard-sphere fluid at density $\rho\sigma^3 = 0.4$. Note that the overlap between the distributions decreases as the chains become longer. The difference $g(\ln \mathcal{W}) - f(\ln \mathcal{W})$ is the overlapping distribution estimated for $\beta\mu^{\text{ex}}$. For the sake of comparison, we also show the value for $\beta\mu^{\text{ex}}$, obtained using the Rosenbluth test particle insertion method (dashed lines).

divide by Z_N and employ the fact that the δ function ensures that $\mathcal{W}_i = \exp(x)$:

$$p_1(x) = e^x \left(\frac{V Z_{\text{id}} Z_N}{Z_{N+1}} \right) \times \frac{\sum_{\text{trials}} \int d\mathbf{Q}^N d\mathbf{s}_i d\{\Gamma_{\text{trial}}\} \exp(-\beta\mathcal{U}(\mathbf{Q}^N)) P_{\text{sel}}(\mathbf{Q}_i) \delta(x - \ln \mathcal{W}_i)}{Z_N}.$$

Finally, we obtain

$$p_1(x) = e^x \frac{V Z_{\text{id}} Z_N}{Z_{N+1}} p_0(x)$$

or

$$\ln p_1(x) = x + \beta\mu_{\text{ex}} + \ln p_0(x).$$

Hence, by constructing a histogram of $\ln w$ both in system 0 (with N chains) and in system 1 (with $N + 1$ chains), we can derive the excess chemical potential of the chain molecules by studying $\ln p_1(x) - \ln p_0(x)$. As in the original Bennett/Shing-Gubbins scheme [341–343], the method works only if there is a range of x values where we have good statistics on both $p_1(x)$ and $p_0(x)$. The advantage of this overlapping distribution scheme over the simple Rosenbluth test particle insertion method is that, with the present method, sampling problems for long chains will manifest themselves as a breakdown of the overlap of p_0 and p_1 . Fig. L.14 shows an example of an application of this overlapping distribution method to hard-sphere polymers.

L.11 Hybrid Monte Carlo

In Molecular Dynamics simulations, all particle coordinates are updated simultaneously. In conventional MC simulations, only a few coordinates are changed in a trial move. As a consequence, collective molecular motions are not well represented by Monte Carlo, and this may adversely affect the rate of equilibration. The advantage of Monte Carlo is that, unlike MD, we can carry out unphysical moves. Moreover, in MC the system is not constrained to move on a hypersurface where some Hamiltonian is conserved. The time step in Molecular Dynamics is limited by the need to conserve energy. Clearly, no such constraint applies to Monte Carlo. For this reason, many authors have attempted to combine the natural dynamics of MD with the large jumps in configuration space possible in MC. The book by Allen and Tildesley [21] describes a number of such techniques (force-bias MC, Langevin Dynamics, smart Monte Carlo) that basically work by including some or all information about the intermolecular forces in the construction of a collective MC trial move.

One scheme that uses MD to generate Monte Carlo trial moves is the so-called hybrid Monte Carlo scheme [564]. At first sight, the advantage of mixing MC and MD is not obvious. However, the criteria for what constitutes a good Monte Carlo trial move are more tolerant than the specifications of a good Molecular Dynamics time step. In particular, one can take a time step that is too long for MD. Energy will not be conserved in such a trial move. However, as long as one uses an algorithm that is time reversible and area-preserving (i.e., that conserves volume in phase space), such collective moves can be used as a Monte Carlo trial move. Fortunately, a systematic way now exists to construct time-reversible, area-preserving MD algorithms, using the multiple-time-step MD scheme of Tuckerman et al. [117]. The usual Metropolis algorithm can then be used to decide on the acceptance or rejection of the move (see, e.g., [565,566]). For every trial move, the particle velocities are chosen at random from a Maxwell distribution. In fact, it is often advantageous to construct a trial move that consists of a sequence of MD steps. The reason is that, due to the randomization of the velocities, the diffusion constant of the system becomes quite low if the velocities are randomized well before the natural decay of the velocity autocorrelation function.

Yet one cannot make the time step for a single hybrid MC move too long, because then the acceptance would become very small. As a consequence, the performance of hybrid MC is not dramatically better than that of the corresponding Molecular Dynamics. Moreover, the acceptance probability of hybrid MC moves of constant length decreases with the system size, because the root-mean-square error in the energy increases with $N^{1/2}$. MD does not suffer from a similar problem. That is to say, the noise in the total energy increases with N , but the stability of the MD algorithm does not deteriorate. Hence, for very large systems, MD will always win. For more normal system sizes, hybrid MC may be advantageous.

It is also interesting to use hybrid MC on models that have an expensive (many-body) potential energy function that may, to a first approximation, be modeled using a cheap (pair) potential. We could then perform a sequence of MD steps, using the cheap potential. At the end of this collective (MD) trial move, we would accept or reject the resulting configuration by applying the Metropolis criterion with the *true* potential energy function. Many variations of this scheme exist. In any event, the hybrid Monte Carlo method is a scheme that requires fine tuning [566].

Forrest and Suter [567] have devised a hybrid MC scheme that samples polymer conformations, using fictitious dynamics of the generalized coordinates. This scheme leads to an improved sampling of the polymer conformations compared to normal MD. The interesting feature of the dynamics used in ref. [567] is that it uses a Hamiltonian that has the same potential energy function as the original polymer model. In contrast, the kinetic part of the Hamiltonian is adjusted to speed up conformational changes.

L.12 General cluster moves

In general, it is not possible to design clusters such that trial moves are always accepted. However, it is often convenient to perform clustering to enhance the acceptance of trial moves. For instance, in molecular systems with very strong short-range attractions, trial moves that pull apart two neighboring particles are very likely to be rejected. It is preferable therefore to include trial moves that attempt to displace the tightly bound particles as a single cluster. To do this, we have to specify a rule for generating clusters. Let us assume that we have such a rule that tells us that particles i and j belong to a single cluster with probability $p(i, j)$ and are disconnected with a probability $1 - p(i, j)$. Here, $p(i, j)$ depends on the state (relative distance, orientation, spin, etc.) of particles i and j . Moreover, we require that $p(i, j)$ be unchanged in a cluster move if both i and j belong to the cluster, and also if neither particle belongs to the cluster. For instance, $p(i, j)$ could depend on the current distance of i and j only. If we denote the potential energy of the old (new) configuration by $\mathcal{U}_0$ ($\mathcal{U}_1$), the detailed balance condition requires that

$$\begin{aligned} \exp(-\beta\mathcal{U}_0) \prod_{kl} [1 - p^f(k, l)] \text{acc}(o \rightarrow n) \\ = \exp(-\beta\mathcal{U}_1) \prod_{kl} [1 - p^r(k, l)] \text{acc}(n \rightarrow o), \end{aligned} \quad (\text{L.12.1})$$

where k denotes a particle *in* the cluster and l a particle outside it. The superscripts f and r denote forward and reverse moves. In writing Eq. (L.12.1), we have assumed that the probability of forming bonds completely within, or completely outside, the cluster is the same for forward and reverse moves. From Eq. (L.12.1), we derive an expression for the ratio of the acceptance probabili-

ties:

$$\frac{\text{acc}(o \rightarrow n)}{\text{acc}(n \rightarrow o)} = \exp\{-\beta[\mathcal{U}(n) - \mathcal{U}(o)]\} \prod_{kl} \frac{1 - p^r(k, l)}{1 - p^f(k, l)}. \quad (\text{L.12.2})$$

Clearly, many choices for p_{kl} are possible. A particularly simple form was chosen by Wu et al. [777], who assumed that $p(i, j) = 1$ for r_{ij} less than a critical distance r_c and $p(i, j) = 0$ beyond that distance (see Example 27). Note that the acceptance rule in Eq. (L.12.1) guarantees that two particles that did not belong to the same cluster in the old configuration will not end up at a distance less than r_c .

Illustration 27 (Micelle formation). Surfactants are amphiphilic molecules that consist of two chemically distinct parts. If these two parts cannot be connected, then one part will preferentially dissolve in another solvent compared to the other part. The most common case is that one part of the molecule is water soluble and the other oil soluble. But, as the two parts of the molecule are connected, the dissolution of surfactant molecules in a pure solvent (say, water) causes frustration, as the oil-soluble part of the molecule is dragged along into the water phase. Beyond a certain critical concentration, the molecules resolve this frustration by self-assembling into micelles. Micelles are (often spherical) aggregates of surfactant molecules in which the surfactants are organized in such a way that the hydrophilic heads point toward the water phase and the hydrophobic tails toward the interior of the micelle. It is of considerable interest to study the equilibrium properties of such a micellar solution. However, Molecular Dynamics simulations on model micelles have shown that the micelles move on a time scale that is long compared to the time it takes individual surfactant molecules to move [778]. Using conventional MC, rather than MD, does not improve the situation: it is relatively easy to move a single surfactant, but it takes many displacements of single surfactants to achieve an appreciable rearrangement of the micelles in the system. Yet, to sample the equilibrium properties of micellar solutions, the micelles must be able to move over distances that are long compared to their own diameter, they must be able to exchange surfactant molecules and they must even be able to merge or break up. As a consequence, standard simulations of micellar self-assembly are very slow.

Wu et al. [777] have used cluster moves to speed up the simulation of micellar solutions. They use a cluster MC scheme that makes it possible to displace entire micelles with respect to each other. The specific model for a surfactant solution that Wu et al. studied was based on a lattice model proposed by Stillinger [779]. In this model, the description of surfactants is highly simplified: the hydrophilic and hydrophobic groups are considered independent (unbonded) particles. The constraint that the head and tail of a surfactant be physically linked is translated into an electrostatic attraction between these groups. The magnitude of these effective charges depends on density and tem-

perature. A typical configuration is shown in Fig. L.15. In such a system it is natural to cluster the molecules in a micelle and subsequently, move entire micelles. The cluster criterion used by Wu et al. is

$$p(i, j) = \begin{cases} 1 & \text{if } r_{ij} < r_c \\ 0 & \text{if } r_{ij} > r_c \end{cases}$$

Wu et al. used $r_c = 1$, which implies that two particles belong to the same cluster if they are on neighboring lattice sites.

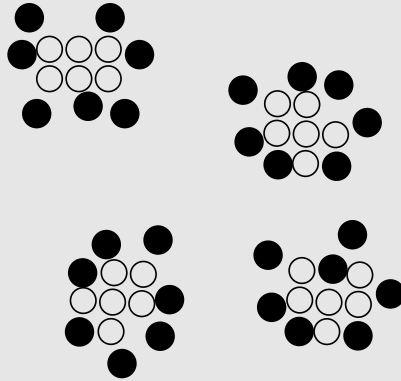


FIGURE L.15 Snapshot of a typical configuration of the surfactant model of Wu et al. [777], in which the hydrophilic and hydrophobic parts of surfactant molecules are represented by charges (black head and white tail). Under appropriate conditions, the surfactants self-assemble into micelles.

In the first step of the algorithm, the clusters are constructed using the preceding criterion. Subsequently, a cluster is selected at random and given a random displacement. It is instructive to consider the case in which we would use the ordinary acceptance rules to move the cluster; that is,

$$\text{acc}(o \rightarrow n) = \min[1, \exp(-\beta \Delta U)].$$

Fig. L.16 shows such a cluster move. The first step (top) is the construction of the cluster, followed by a displacement of one of the clusters (middle). If we accept this move with the probability given by the preceding equation, we would violate microscopic reversibility. Since we have moved the clusters in such a way that they touch each other, in the next step, these two clusters would be considered a single cluster. It then will be impossible to separate them to retrieve the initial configuration.

If we use the correct acceptance rule, Eq. (L.12.2),

$$\text{acc}(o \rightarrow n) = \min \left[1, \exp(-\beta \Delta U) \prod_{kl} \frac{1 - p^{\text{new}}(k, l)}{1 - p^{\text{old}}(k, l)} \right],$$

then this move would be rejected because $p^{\text{new}}(k, l) = 1$. Since these cluster moves do not change the configuration of the particles in a cluster and do not change the total number of particles in a cluster, it is important to combine these cluster moves with single particle moves, or use a cluster criterion $p(i, j)$ that allows the number of particles in a cluster to change.

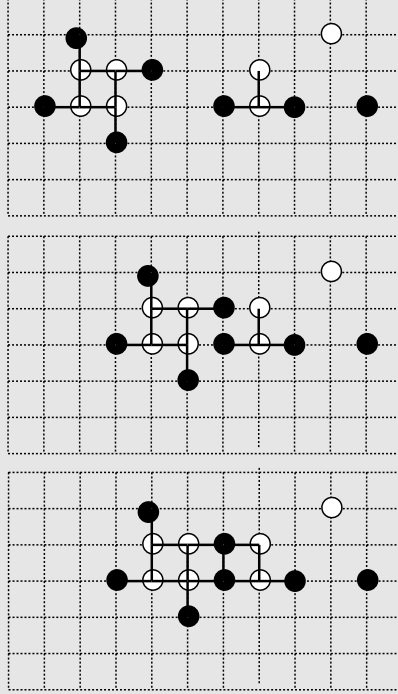


FIGURE L.16 Violation of detailed balance in a cluster move; the top figure shows the four clusters in the system; in the middle figure one of the clusters is given a random displacement, which brings this cluster into contact with another cluster; and the bottom figure shows that the new configuration has only three clusters if the moves have been accepted.

Orkoulas and Panagiotopoulos have used such cluster moves to simulate the vapor-liquid coexistence curve of the restricted primitive model of an ionic fluid [780,781].

L.13 Boltzmann-sampling with dissipative particle dynamics

The original version of **DPD** [257] did not generate a Boltzmann distribution. Español and Warren [683] corrected this problem. In the analysis of ref. [683], the time evolution generated by a **DPD** algorithm is written in the form of a Fokker-Planck equation (see e.g., [67]):

$$\partial_t \mathcal{N}(r, p; t) = \mathcal{L}_C \mathcal{N}(r, p; t) + \mathcal{L}_D \mathcal{N}(r, p; t), \quad (\text{L.13.1})$$

where $\mathcal{L}_C$ is the usual Liouville operator of a Hamiltonian system interacting with conservative forces $\mathbf{F}^C$,

$$\mathcal{L}_C \equiv - \left[\sum_i \frac{\mathbf{p}_i}{m} \frac{\partial}{\partial \mathbf{r}_i} + \sum_{i,j} \mathbf{f}_{ij}^C \frac{\partial}{\partial \mathbf{p}_i} \right], \quad (\text{L.13.2})$$

and the operator $\mathcal{L}_D$ takes into account the effects of the dissipative and random forces:

$$\mathcal{L}_D \equiv \sum_{i,j} \hat{\mathbf{r}}_{ij} \frac{\partial}{\partial \mathbf{p}_i} \left[\gamma \omega_D(r_{ij}) (\hat{\mathbf{r}}_{ij} \cdot \mathbf{v}_{ij}) + \sigma^2 \omega_R^2(r_{ij}) \hat{\mathbf{r}}_{ij} \left(\frac{\partial}{\partial \mathbf{p}_i} - \frac{\partial}{\partial \mathbf{p}_j} \right) \right]. \quad (\text{L.13.3})$$

The derivation of these equations uses techniques developed for stochastic differential equations. The advantage of casting the [DPD](#) equations in a Fokker-Planck form is that the steady-state solution of Eq. [\(L.13.1\)](#) then corresponds to

$$\partial_t \mathcal{N}_{\text{eq}}(r, p; t) = 0.$$

To make the connection with statistical mechanics, the steady-state solution should correspond to the canonical distribution:

$$\begin{aligned} \mathcal{N}_{\text{eq}}(r, p; t) &= \frac{1}{Q_{NVT}} \exp[-\beta \mathcal{H}(r, p)] \\ &= \frac{1}{Q_{NVT}} \exp \left[-\beta \left(\sum_i p_i^2 / 2m_i + V(r) \right) \right], \end{aligned}$$

where $V(r)$ is the potential that gives rise to the conservative forces, $\mathcal{H}(r, p)$ is the Hamiltonian, and Q_{NVT} is the partition function of the NVT ensemble. By construction, this Boltzmann distribution satisfies

$$\mathcal{L}_C \mathcal{N}_{\text{eq}}(r, p; t) = 0.$$

We, therefore, need to ensure that

$$\mathcal{L}_D \mathcal{N}_{\text{eq}}(r, p; t) = 0.$$

This is achieved by imposing that

$$\omega_R^2(r) = \omega_D(r) \quad \text{and} \quad \sigma^2 = 2k_B T \gamma,$$

which are Eqs. [\(16.1.4\)](#) and [\(16.1.5\)](#), i.e., the choice made by Español and Warren [\[683\]](#).

L.14 Reference states

L.14.1 Grand-canonical ensemble simulation

In a grand-canonical ensemble simulation, we impose the temperature and chemical potential. Experimentally, however, usually, the pressure rather than the chemical potential of the reservoir is fixed. To compare the experimental data with the simulation results it is necessary therefore to determine the pressure that corresponds to a given value of the chemical potential and temperature of our reservoir.

L.14.1.1 Preliminaries

The partition function of a system with N atoms in the N, V, T -ensemble is given by

$$Q(N, V, T) = \frac{V^N}{\Lambda^{3N} N!} \int d\mathbf{s}^N \exp[-\beta \mathcal{U}(\mathbf{s}^N)], \quad (\text{L.14.1})$$

where $\mathbf{s}^N$ are the scaled coordinates of the N particles. The free energy is related to the partition function via

$$F = -\frac{1}{\beta} \ln Q(N, V, T),$$

which gives us for the chemical potential

$$\mu \equiv \frac{\partial F}{\partial N} = -\frac{1}{\beta} \ln[Q(N+1, V, T)/Q(N, V, T)]. \quad (\text{L.14.2})$$

For a system consisting of N molecules with each molecule having M atoms, the partition function is

$$Q(N, M, V, T) = \frac{q(T)^N V^N}{N!} \prod_{i=1}^M \int d\mathbf{s}_i^N \exp[-\beta \mathcal{U}(\mathbf{s}_i^N)], \quad (\text{L.14.3})$$

where $q(T)$ is the part of the partition function of a molecule that contains the integration over momenta (for an atom, $q(T)$ is simply Λ^{-3}) and the $\mathbf{s}^N$ are the Cartesian coordinates of atoms in the molecule. It should be stressed that, in writing Eq. (L.14.3), we are making the assumption that there are no “hard” constraints on these intramolecular coordinates. In the presence of hard constraints, the integral in Eq. (L.14.3) would contain a Jacobian (see section 14.1).

L.14.1.2 Ideal gas

In the limit of zero density, any system will behave as an ideal gas. In this limit, only the intramolecular interactions contribute to the total potential energy

$$\mathcal{U} \approx \sum_{i=1}^N \mathcal{U}^{\text{intra}}(i).$$

For a system consisting of noninteracting atoms, the partition function (L.14.1) reduces to

$$Q_{\text{IG}}(N, V, T) = \frac{V^N}{\Lambda^{3N} N!}. \quad (\text{L.14.4})$$

We can write, for the chemical potential of such an ideal gas of atoms,

$$\mu_{\text{id.gas}} = \mu_{\text{id.gas}}^0 + k_B T \ln \rho, \quad (\text{L.14.5})$$

with the chemical potential of the reference state defined by

$$\mu_{\text{id.gas}}^0 \equiv k_B T \ln \Lambda^3. \quad (\text{L.14.6})$$

In the case of gas of noninteracting molecules, the partition function (L.14.3) reduces to

$$Q_{\text{id.gas}}(N, M, V, T) = \frac{q(T)^N V^N}{N!} \left\{ \prod_{i=1}^M \int d\mathbf{s}_i \exp[-\beta \mathcal{U}^{\text{intra}}(\mathbf{s}_i)] \right\}^N. \quad (\text{L.14.7})$$

Substitution into Eq. (L.14.2) yields, for the chemical potential,

$$\mu_{\text{id.gas}} = \mu_{\text{id.gas}}^0 + k_B T \ln \rho, \quad (\text{L.14.8})$$

where the reference chemical potential is defined as

$$\begin{aligned} \beta \mu_{\text{id.gas}}^0 &\equiv -\ln q(T) + \beta \mu_{\text{intra}}^0 \\ &= -\ln q(T) - \ln \left[\prod_{i=1}^M \int d\mathbf{s}_i \exp[-\beta \mathcal{U}^{\text{intra}}(\mathbf{s}_i)] \right]. \end{aligned} \quad (\text{L.14.9})$$

Note that $\mu_{\text{id.gas}}^0$ depends only on temperature. At any given temperature, it simply acts as a constant shift of the chemical potential that has no effect on the observable thermodynamic properties of the system.

L.14.1.3 Grand-canonical simulations

In a grand-canonical simulation, we use the following acceptance rules (see section 6.5.2). In addition, we have

$$\begin{aligned} \text{acc}(N \rightarrow N+1) \\ = \min \left[1, \frac{Vq(T) \exp(-\beta\mu_{\text{intra}}^0)}{(N+1)} \exp\{\beta[\mu^B - \mathcal{U}(N+1) + \mathcal{U}(N)]\} \right]. \end{aligned}$$

For the removal of a particle, we have

$$\begin{aligned} \text{acc}(N \rightarrow N-1) \\ = \min \left[1, \frac{N}{q(T) \exp(-\beta\mu_{\text{intra}}^0)V} \exp\{-\beta[\mu^B + \mathcal{U}(N-1) - \mathcal{U}(N)]\} \right]. \end{aligned}$$

These equations are based on the idea that particles are exchanged with a reservoir containing the same molecules at the same chemical potential, the only difference being that, in the reservoir, the molecules do not interact. In practical cases (e.g., adsorption), this means that we have a dense phase in equilibrium with a dilute vapor. And, whereas the absolute chemical potential of the vapor is of little interest, the absolute pressure is clearly an important quantity. The pressure in the reservoir is related to the chemical potential through

$$\begin{aligned} \beta\mu^B &\equiv \beta\mu_{\text{id.gas}}^0 + \ln(\rho) \\ &= \beta\mu_{\text{id.gas}}^0 + \ln(\beta P_{\text{id.gas}}). \end{aligned} \quad (\text{L.14.10})$$

Substitution of this expression in the acceptance rules yields

$$\text{acc}(N \rightarrow N+1) = \min \left[1, \frac{V\beta P_{\text{id.gas}}}{(N+1)} \exp\{-\beta[\mathcal{U}(N+1) - \mathcal{U}(N)]\} \right] \quad (\text{L.14.11})$$

for the addition of a particle and a similar expression for particle removal. In other words, if the experimental conditions are such that the system of interest is in equilibrium with a reservoir that behaves like an ideal gas, then only the pressure of this effectively ideal gas enters into the acceptance rules for trial moves. All information about the reference state drops out (as expected).

If the pressure in the reservoir is too high for the ideal gas law to hold, we have to use an equation of state to relate the chemical potential of the reservoir to its pressure:

$$\beta\mu_B = \beta\mu_{\text{id.gas}}^0 + \ln(\beta P\phi), \quad (\text{L.14.12})$$

where ϕ is the fugacity coefficient of the fluid in the reservoir. The fugacity coefficient can be computed directly from the equation of state of the vapor in the reservoir. It is important to note that this fugacity coefficient is a function of the temperature and pressure. In summary, for a non-ideal gas, we should replace $P_{\text{id.gas}}$ in the acceptance rule (L.14.11) by $P\phi$.